Copyrighted Materials
Copyright © 2002 Oxford University Press Retrieved from www.knovel.com

3

Nonsinusoidal Excitations and Impact

3.1 INTRODUCTION

This chapter addresses the problem of how to find a system's response when the input excitation is not a simple sinusoid. If you think about it, it seems pretty obvious that almost no real system will have a purely sinusoidal input. Many, many of the inputs will be periodic; i.e., the input will be made up of an endlessly repeating waveform. But the chances that the waveform is a pure sine wave are pretty low. For instance, the repeated firings inside the cylinder of an internal combustion engine will produce a periodic, but not sinusoidal, loading on the piston. Thus we're motivated to figure out how we can determine a system's steady state response for these cases.

Furthermore, the input to our system might not even be periodic. Oftentimes the loading will be very transient in nature. An earthquake is a good example of a transient loading. The ground will shake for a few seconds and then stop. Our problem then consists of determining a structure's response to a loading that exists only for a finite amount of time. If the earthquake lasts for a long time, then we might actually view the problem in a steady state way. But if the earthquake is of short duration, then a transient viewpoint is more appropriate. A similar problem will

occur when an airplane encounters a momentary updraft. The updraft will load the wings for only a short time, and the wings will then bend (and the airplane will displace), in response to the input.

The loading might even be *very* transient. This would be the case for impact, such as the one shown in the chapter's opening cartoon. Impacts and explosions will create high levels of force that last for only a very short time. Dropping an object onto the ground is one way to create a severe impact, and we'll definitely want to know how to handle such cases if we want our designs to withstand normal usage. In industry, this type of analysis affects packaging and shipping. The manufacturer wants the product to withstand the normal operating environment and also to withstand the loads that may occur prior to installation. Interestingly, some of the most severe loadings applied to automobiles happen, not during normal driving but during shipment to dealers. Automobile manufacturers, who spend a good deal of effort to determine the nonsinusoidal loads vehicles will experience during shipment, work hard to ensure that they survive.

The final type of signal we'll consider, the random signal, falls between the first two types. Periodic signals have a repeating pattern that doesn't die down. Transient signals have no periodicity and a finite lifetime. Random signals are not periodic but don't die down. Random excitations are very common in real life. The force of wind on a building will be random. There's always some wind, but the direction and force vary randomly. The vibration of a floor in a manufacturing facility will be random: the inputs due to people walking about, parts being dropped or moved occasionally, doors being open and closed, etc. If an earthquake goes on for an extended time, its motions can be viewed as random.

We'll start the chapter by looking at the most immediately accessible extension to what we already know, the response of a vibratory system to a periodic, but not sinusoidal, input. Following this, we'll consider what needs to be done to deal with transient inputs. Finally, we'll look at the problem of random loading.

3.2 FOURIER SERIES ANALYSIS

This section will cover how to analyze systems for which the excitation is periodic but not made up of a single sinusoid. We've mentioned that superposition lets us easily determine the output of a system to an input $y_1 + y_2$ if we already know the corresponding outputs x_1 and x_2 for the individual inputs (it's simply equal to $x_1 + x_2$). Just to clarify this idea a bit more, let's work out a particular example. Say we want to know the response of a seismically excited system for the input

$$y(t) = \frac{8}{\pi^2}\sin(\omega t) - \frac{32}{36\pi^2}\sin(3\omega t) \tag{3.2.1}$$

A plot of $y(t)$ versus time is shown in Figure 3.1. Note that the addition of the $\sin(3\omega t)$ term has served to sharpen up the base sinusoid, making it more triangular in shape. This result isn't a coincidence, and it's due to the particular choice of amplitudes used. We'll look at this more in a few pages.

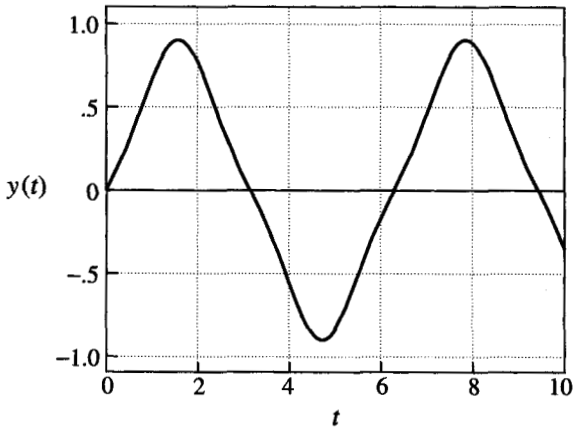


Figure 3.1 Sum of two sinusoids

The governing equation of motion (2.3.3) is repeated here for convenience:

$$\ddot{x} + \omega_n^2 x = \omega_n^2 y \tag{3.2.2}$$

Since the input consists of two, linearly independent excitations, and the dynamical system has no damping, we can assume a response

$$x(t) = b_1 \sin(\omega t) + b_2 \sin(3\omega t) \tag{3.2.3}$$

which, when differentiated twice and substituted into (3.2.2), leads to

$$b_1(\omega_n^2 - \omega^2)\sin(\omega t) + b_2(\omega_n^2 - 9\omega^2)\sin(3\omega t) = \frac{8\omega_n^2}{\pi^2}\sin(\omega t) - \frac{32\omega_n^2}{36\pi^2}\sin(3\omega t) \tag{3.2.4}$$

Since $\sin(3\omega t)$ and $\sin(\omega t)$ are linearly independent, (3.2.4) can be satisfied only if the coefficients of each sinusoid on the right and left sides of the equation are equal to each other, or

$$b_1(\omega_n^2 - \omega^2) = \frac{8\omega_n^2}{\pi^2} \tag{3.2.5}$$

and

$$b_2(\omega_n^2 - 9\omega^2) = -\frac{32\omega_n^2}{36\pi^2} \tag{3.2.6}$$

Thus we can easily solve for b_1 and b_2:

$$b_1 = \frac{8}{\pi^2}\frac{\omega_n^2}{\omega_n^2 - \omega^2} \tag{3.2.7}$$

$$b_2 = -\frac{32}{36\pi^2}\frac{\omega_n^2}{\omega_n^2 - 9\omega^2} \tag{3.2.8}$$

Therefore

$$x(t) = \frac{8}{\pi^2}\frac{\omega_n^2}{\omega_n^2 - \omega^2}\sin(\omega t) - \frac{32}{36\pi^2}\frac{\omega_n^2}{\omega_n^2 - 9\omega^2}\sin(3\omega t) \tag{3.2.9}$$

As advertised, this is exactly the same result as if we had solved

$$\ddot{x} + \omega_n^2 x = \frac{8\omega_n^2}{\pi^2}\sin(\omega t) \tag{3.2.10}$$

to obtain

$$x(t) = \frac{8}{\pi^2}\frac{\omega_n^2}{\omega_n^2 - \omega^2}\sin(\omega t) \tag{3.2.11}$$

and had then solved

$$\ddot{x} + \omega_n^2 x = \frac{32}{36\pi^2}\sin(3\omega t) \tag{3.2.12}$$

to obtain

$$x(t) = -\frac{32}{36\pi^2}\frac{\omega_n^2}{\omega_n^2 - 9\omega^2}\sin(3\omega t) \tag{3.2.13}$$

and added the two solutions together. Note that the solutions have been written in the form

$$\text{(amplitude of forcing)(transfer function)(time variation of forcing)} \tag{3.2.14}$$

This is to make it doubly clear that the response of a linear system is just the transfer function multiplied by the excitation.

Now we can begin to see why superposition is such a powerful concept. It implies that we can easily find the response of dynamical systems as long as we can consider the inputs to be made up of individual sinusoids. Thus the obvious question is: when can we view an input to a system as being made up of individual sinusoidal components? Clearly $y(t) = e^{-2t}$ isn't a likely candidate, since it is patently nonsinusoidal. But what about a triangular input, like that illustrated in Figure 3.2? Is this made up of sinusoids? We might be tempted to say yes, since this figure looks a good bit like Figure 3.1, which we've just seen is composed of $\sin(\omega t)$ and $\sin(3\omega t)$. In fact, Jean Baptiste Joseph Fourier, in the early 1800s, showed that one can approximate any well-behaved, periodic signal as the infinite sum of a series of pure sinusoids. Note the *periodic*. At this point we will consider only inputs that are periodic; i.e., if $f(t)$ is the function under consideration, then $f(t+T) = f(t)$, where T is the period of the function. Later in the chapter we'll consider what can be done when the functions of interest are not periodic.

According to Fourier, we should be able to represent a periodic signal, $f(t)$, as

$$f(t) = \sum_{n=0}^{\infty} a_n \cos(n\omega_0 t) + \sum_{n=1}^{\infty} b_n \sin(n\omega_0 t) \tag{3.2.15}$$

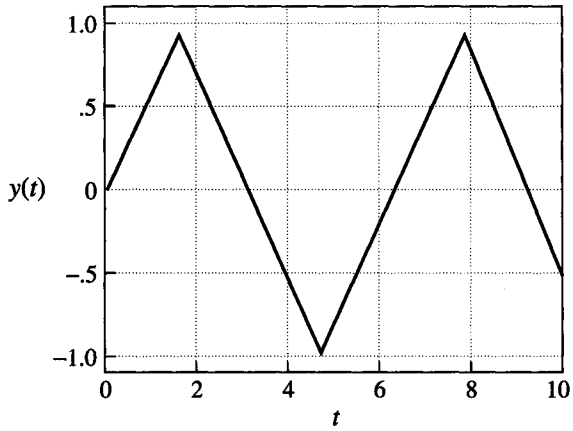


Figure 3.2 Triangle wave

The ω_0 component is the lowest frequency (longest wavelength) present in the signal $f(t)$ and is related to the signal's period T by

$$\omega_0 = \frac{2\pi}{T} \tag{3.2.16}$$

You'll note that the summations in the preceding equation go to infinity. This means that if we *really* wanted to precisely model an arbitrary periodic signal, we might need an infinite number of sinusoids. Luckily, we will never need an infinite number of terms for our analyses, for reasons that will become clear momentarily.

Assuming that we can indeed approximate $f(t)$ by a series of sinusoids, the problem still remains of how to determine a_n and b_n, the coefficients that multiply the sinuosoids. To make the general procedure clear, let's first work out a specific example. The function that we are going to analyze is

$$f_1(t) = 2\sin(\omega_0 t) - 3\cos(2\omega_0 t) \tag{3.2.17}$$

Thus we have

$$\sum_{n=0}^{\infty} a_n \cos(n\omega_0 t) + \sum_{n=1}^{\infty} b_n \sin(n\omega_0 t) = f_1(t) = 2\sin(\omega_0 t) - 3\cos(2\omega_0 t) \tag{3.2.18}$$

Of course, it's obvious that we can just read off the correct answers ($a_2 = -3$, $b_1 = 2$, and all other a_n and $b_n = 0$), but for the purpose of this discussion we'll assume that we don't know the answers in this way, that we'll have to calculate them from $f(t)$. This procedure is what we'd actually have to do in a real Fourier series analysis, one in which we didn't know the sinusoidal contributions beforehand.

Here's how we'll do it. First, multiply both sides of (3.2.18) by $\sin(m\omega_0 t)$. This is the sinusoid associated with b_m. For our particular example we'll expect answers only for $m = 1$, since that's the

only sine component in $f(t)$. This gives us

$$\sum_{n=0}^{\infty} a_n \cos(n\omega_0 t)\sin(m\omega_0 t) + \sum_{n=1}^{\infty} b_n \sin(n\omega_0 t)\sin(m\omega_0 t)$$
$$= 2\sin(\omega_0 t)\sin(m\omega_0 t) - 3\cos(2\omega_0 t)\sin(m\omega_0 t) \tag{3.2.19}$$

We can simplify matters a bit by reexpressing these products of sines and cosines. Recall that

$$\sin(m\omega_0 t) = \frac{1}{2i}(e^{im\omega_0 t} - e^{-im\omega_0 t}) \tag{3.2.20}$$

and

$$\cos(m\omega_0 t) = \frac{1}{2}(e^{im\omega_0 t} + e^{-im\omega_0 t}) \tag{3.2.21}$$

Using (3.2.20) and (3.2.21), we can expand (3.2.19) out into its exponential parts and then combine these exponentials back into sines and cosines:

$$\sum_{n=0}^{\infty} a_n \left(\frac{1}{2}\right)\left[\sin\big((n+m)\omega_0 t\big) - \sin\big((n-m)\omega_0 t\big)\right]$$
$$+ \sum_{n=1}^{\infty} b_n \left(\frac{1}{2}\right)\left[-\cos\big((n+m)\omega_0 t\big) + \cos\big((n-m)\omega_0 t\big)\right]$$
$$= 2\left(\frac{1}{2}\right)\left[-\cos\big((1+m)\omega_0 t\big) + \cos\big((1-m)\omega_0 t\big)\right]$$
$$- 3\left(\frac{1}{2}\right)\left[\sin\big((2+m)\omega_0 t\big) - \sin\big((2-m)\omega_0 t\big)\right] \tag{3.2.22}$$

Next we're going to exploit a property called *orthogonality*, something we haven't talked about yet, but it will be easier to understand once you see it at work. To make use of orthogonality, we will integrate both sides of (3.2.22) from $t = 0$ to $t = T$, i.e. over one period. So we'll have

$$\sum_{n=0}^{\infty} a_n \left(\frac{1}{2}\right)\left[\int_0^{\frac{2\pi}{\omega_0}} \sin\big((n+m)\omega_0 t\big)\,dt - \int_0^{\frac{2\pi}{\omega_0}} \sin\big((n-m)\omega_0 t\big)dt\right]$$
$$+ \sum_{n=1}^{\infty} b_n \left(\frac{1}{2}\right)\left[-\int_0^{\frac{2\pi}{\omega_0}} \cos\big((n+m)\omega_0 t\big)\,dt + \int_0^{\frac{2\pi}{\omega_0}} \cos\big((n-m)\omega_0 t\big)dt\right]$$
$$= 2\left(\frac{1}{2}\right)\left[-\int_0^{\frac{2\pi}{\omega_0}} \cos\big((1+m)\omega_0 t\big)\,dt + \int_0^{\frac{2\pi}{\omega_0}} \cos\big((1-m)\omega_0 t\big)\,dt\right]$$
$$- 3\left(\frac{1}{2}\right)\left[\int_0^{\frac{2\pi}{\omega_0}} \sin\big((2+m)\omega_0 t\big)\,dt + \int_0^{\frac{2\pi}{\omega_0}} \sin\big((2-m)\omega_0 t\big)\,dt\right] \tag{3.2.23}$$

Taken all together, this looks *really* messy. However, if we look at each term individually we notice a remarkable thing. It seems that almost every term is equal to zero since, for each term, all we have is an integration over one or more periods of a pure sinusoid. And we know that integrating a cosine or sine over one or more periods yields zero as a result. For instance, if $n = 1$ and $m = 5$, then the first integral of the equation is

$$\int_0^{\frac{2\pi}{\omega_0}} \sin(6\omega_0 t)\, dt = -\frac{\cos(6\omega_0 t)\Big|_0^{\frac{2\pi}{\omega_0}}}{6\omega_0} = 0 \tag{3.2.24}$$

This will be true for any combination of n and m except those for which $n = m$. If $n = m$, then the $\sin((n - m)\omega_o t)$ term equals zero. Thus it isn't a finite sinusoid anymore. However it's equal to zero and so gives us no trouble. However $\cos((n - m)\omega_0 t)$ goes to 1. Therefore the fourth integral on the left-hand side of (3.2.23) becomes

$$\int_0^{\frac{2\pi}{\omega_0}} dt = \frac{2\pi}{\omega_0} \tag{3.2.25}$$

Let's take a second to recall what we've found. We already know that only two harmonics show up in our signal, namely, $2\sin(\omega_0 t)$ and $-3\cos(2\omega_0 t)$. We formed a general Fourier series, set it equal to our signal, multiplied by $\sin(m\omega_0 t)$, and integrated over a cycle. We then found that almost every term went to zero. If m isn't equal to 1, then *all* the terms on the right-hand side of (3.2.23) are equal to zero. Only if $m = 1$ do we obtain a finite result.

If we choose $m = 1$, then (3.2.23) becomes

$$b_1 \frac{\pi}{\omega_0} = \frac{2\pi}{\omega_0} \tag{3.2.26}$$

or

$$b_1 = 2 \tag{3.2.27}$$

This is the correct result! Of course, we knew when we started that the coefficient of the Fourier series for $\sin(\omega_0 t)$ had to be 2 because that's how much of that particular harmonic was in the signal originally. The point is that we usually *don't* know ahead of time how much of a particular harmonic there will be, and the procedure we've just gone over will give us the answer. We could go ahead and find the cosine harmonics in the same way; simply multiply by $\cos(m\omega_0 t)$, integrate over a period, and vary m until we get a finite answer. The general procedure can be written as

$$a_n = \frac{\omega_0}{\pi} \int_0^{\frac{2\pi}{\omega_0}} f(t)\cos(n\omega_0 t)dt \tag{3.2.28}$$

and

$$b_n = \frac{\omega_0}{\pi} \int_0^{\frac{2\pi}{\omega_0}} f(t)\sin(n\omega_0 t)dt \tag{3.2.29}$$

The only other twist you'll need to remember is that the signal you're analyzing may not average out to zero. In such a case we say it has a bias or a constant offset. Since constants are even functions, we can view this as a cosine with zero frequency and so we'll have

$$a_0 = \frac{\omega_0}{2\pi} \int_0^{\frac{2\pi}{\omega_0}} f(t)dt \tag{3.2.30}$$

These coefficients are then used in

$$f(t) = \sum_{n=0}^{\infty} a_n \cos(n\omega_0 t) + \sum_{n=1}^{\infty} b_n \sin(n\omega_0 t) \tag{3.2.31}$$

to define our Fourier series.

You're probably thinking that this looks a bit tedious to do by hand. And it is. But it's a snap to do on a computer. You just need a short subroutine that takes in a signal and then decomposes it into its individual harmonics, using pretty much the same approach that we've just gone over.

Please don't dismiss all this lightly. The ideas behind the Fourier series will pop up again and again in the future. What we're doing is using (3.2.28)–(3.2.30) as *filters*. The indicated integration operations really do sift through $f(t)$ and filter out everything except the harmonic we're looking for. If that harmonic is in the signal, then these calculations will find it. This is a very powerful idea and finds uses not only in linear vibrations but in nonlinear vibrations as well.

By determining the Fourier series representation of the signal, we have done a very significant thing: we have moved from a time domain description into a frequency domain. We now can determine how much of a certain frequency is present in a given signal. This kind of a viewpoint is *very* prevalent in a wide range of applications. For instance, a great deal of control work is done almost entirely in the frequency domain. The control designer specifically looks for control strategies that will reduce the amount of motion in a system at specific frequencies and is willing to let the motion increase at other frequencies to achieve this. A frequency viewpoint also is useful in connection with the Tacoma-Narrows bridge disaster. Most of you saw film clips of the Tacoma-Narrows collapse in high school. For those who didn't, this spectacular bridge failure, several decades ago, occurred as a result of its own excessive motions. (Really excessive!) The Tacoma-Narrows bridge failed because the designers hadn't considered what sort of dynamic responses the bridge might have under different wind conditions. If they had known what to look for, they would have realized that a particular type of motion, occurring at a particular frequency, could be excited by certain wind conditions and that the wind would pump energy into the bridge's motion until the structure was destroyed (which is what occurred). The fundamental behavior is one that is most easily viewed from a frequency standpoint.

Finally, almost all studies of structural systems are concerned with the frequency content of the system. The reason for this is that periodic excitations close to the system's natural response frequencies can cause extremely large deflections of the system, deflections that can threaten the structural integrity of the system.

We'll examine spectral representations more fully later in the chapter. For now we will have to be content with knowing that any periodic forcing of our system can be broken down into several

sinusoidal inputs and that we can therefore determine the response of our system by first determining the individual responses to all the separate harmonic components of the forcing and then adding them together, a process that is graphically shown in Figure 3.3. At the top we see a complex periodic input broken up into its constituent harmonic parts by means of a Fourier series analysis. Next, each harmonic is fed through the system being analyzed. As a result, the amplitude and phase of each component are altered. Finally, the resultant outputs are summed into the final response. Note that the input looks like a single harmonic with some superimposed higher harmonic "wiggles", while the output is much more irregular. This change occurred because resonance (due to the system's low damping) greatly amplified the harmonic near the system's natural frequency. Thus we see that a low magnitude component of the input signal can become significant or even dominant in the output.

It's worth reemphasizing that this Fourier series analysis technique is one of the powerful results that comes of having a linear system. Only when the governing differential equation is linear (all dependent variables and their derivatives appear only to the first power) can we break up a signal, calculate the individual responses, add them up, and end up with the actual solution. In a nonlinear system this would *not* work.

Example 3.1

Problem Determine the response of the system shown in Figure 3.4 when the base motion is in the form of the triangle wave illustrated.

Solution Although we could go ahead and just "plug and chug" using (3.2.28)–(3.2.31), we'll spend some time thinking about the problem and see how doing so can sometimes reduce our work level. The first thing to notice about the input forcing is that it is an odd function. Therefore we know immediately that all the a_n coefficients in (3.2.31) are going to equal zero. (If the function is odd, then it can't contain any even components.) Next, we can realize that only the odd harmonics ($n = 1, 3, 5, \ldots$) will show up because of the function's symmetry. Figure 3.5 shows why this is so. If an even harmonic ($n = 2, 4, \ldots$) was present, it would skew the function, making it look lopsided. Odd harmonics won't disturb the function's symmetry. Thus we need concern ourselves only with the odd harmonics. Furthermore, when we multiply the function by $\sin(n\omega_0 t)$ and integrate, we'll get an equal contribution from the section going from 0 to $\frac{\pi}{\omega_0}$ as we do from the section between $\frac{\pi}{\omega_0}$ to $\frac{2\pi}{\omega_0}$. Thus we can calculate the integral over just one interval and double the answer. And finally, we can notice that the same argument holds for our remaining interval. The integral from 0 to $\frac{\pi}{2\omega_0}$ is the same as that from $\frac{\pi}{2\omega_0}$ to $\frac{\pi}{\omega_0}$. Therefore we need only calculate the integral from 0 to $\frac{\pi}{2\omega_0}$ and multiply the answer by 4. Figure 3.6 shows this. The sectors labeled A all contain the same area.

Based on this analysis, our problem has simplified down to solving

$$b_n = 4\frac{\omega_0}{\pi}\int_0^{\frac{\pi}{2\omega_0}} \sin(n\omega_0 t)\left(\frac{2a\omega_0 t}{\pi}\right)dt, \quad n = 1, 3, 5, \ldots$$

This isn't actually all that bad to solve. If we integrate by parts, we'll have

$$b_n = \frac{8a\omega_0^2}{\pi^2}\left(-\frac{t\cos(n\omega_0 t)}{n\omega_0}\bigg|_0^{\frac{\pi}{2\omega_0}} + \int_0^{\frac{\pi}{2\omega_0}} \frac{\cos(n\omega_0 t)}{n\omega_0}\,dt\right), \quad n = 1, 3, 5, \ldots.$$

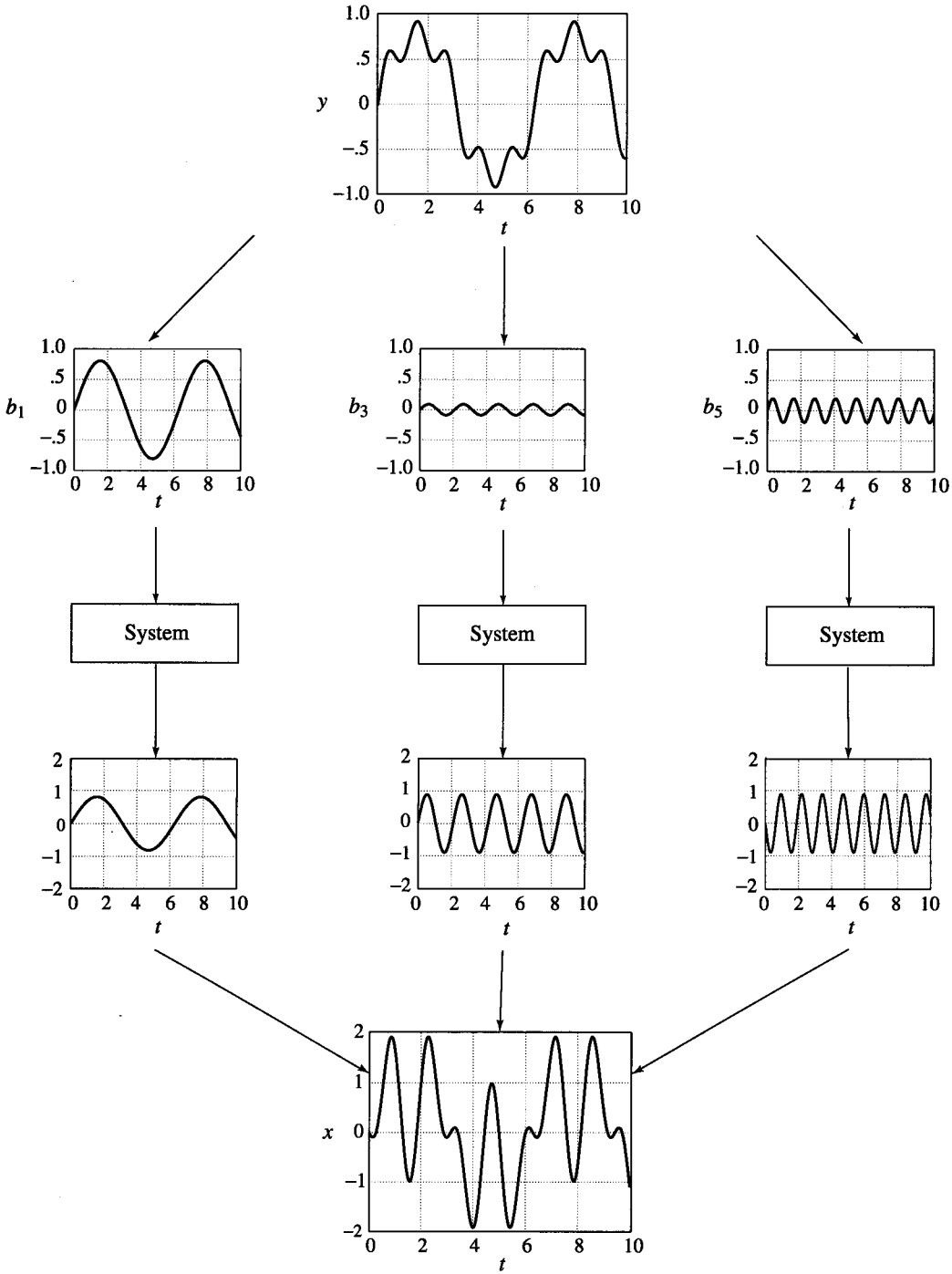


Figure 3.3 Schematic of how Fourier analysis works

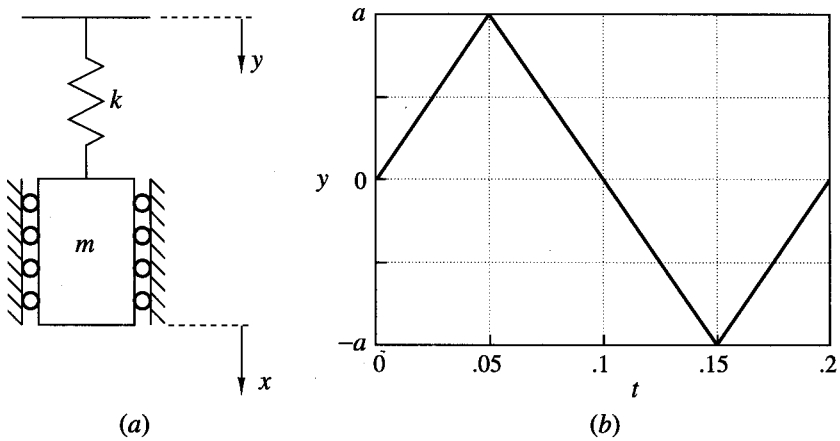


Figure 3.4 Seismically excited system, triangle wave input

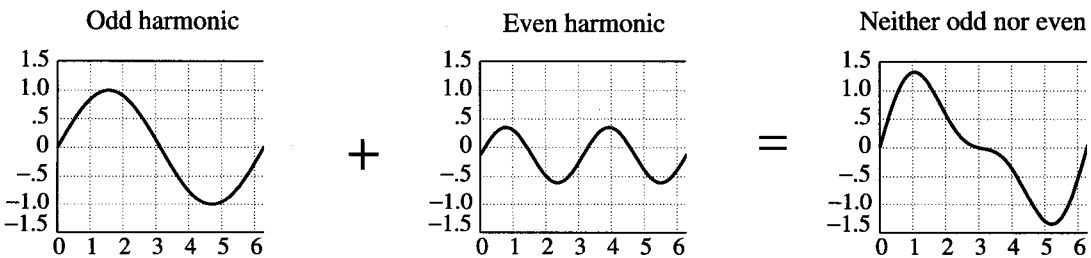


Figure 3.5 Adding odd- and even-order harmonics destroys symmetry

The first term is always zero for odd values of n, and the second term integrates out to

$$\frac{8a\omega_0^2}{\pi^2}\left(\frac{\sin(n\omega_0 t)}{(n\omega_0)^2}\bigg|_0^{\frac{\pi}{2\omega_0}}\right), \quad n = 1, 3, 5, \ldots$$

The sine can only yield 1.0 or -1.0, depending on the value of n, and so the final solution becomes

$$b_{2n-1} = \frac{8a}{\pi^2}\frac{(-1)^{n+1}}{(2n-1)^2}, \quad n = 1, 2, \ldots$$

Now that we've got the b_n coefficients, we can put them together to obtain our Fourier series representation of the triangle wave:

$$y(t) = \frac{8a}{\pi^2}\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{(2n-1)^2}\sin\left((2n-1)\omega_0 t\right), \quad n = 1, 2, \ldots$$

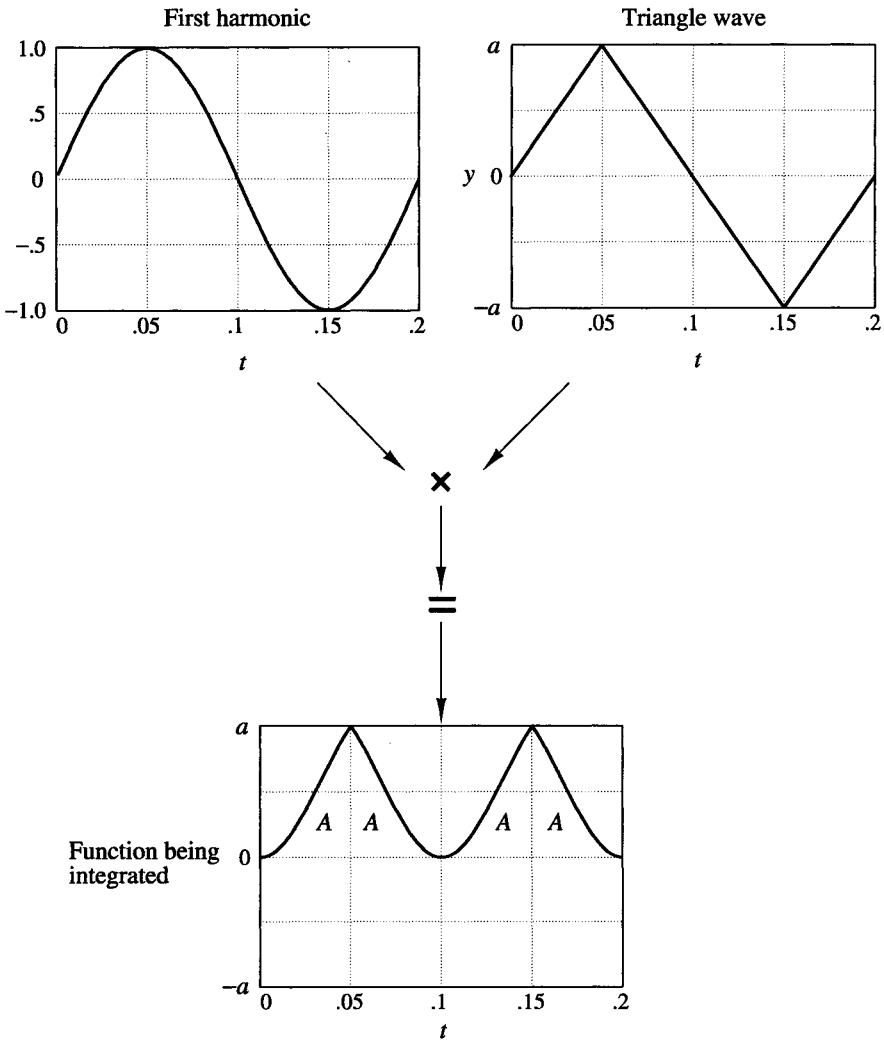


Figure 3.6 Symmetry in the integrals of a particular Fourier decomposition

The rest is reasonably straightforward. We know that the transfer function for the illustrated system is

$$g(\omega) = \frac{\omega_n^2}{\omega_n^2 - \omega^2}$$

Therefore we can immediately write the complete response by combining the two preceding equations to obtain

$$x(t) = \frac{8a}{\pi^2} \sum_{n=1}^{\infty} \left(\frac{(-1)^{n+1}}{(2n-1)^2} \right) \left(\frac{\omega_n^2}{\omega_n^2 - (2n-1)^2 \omega_0^2} \right) \sin\left((2n-1)\omega_0 t\right), \quad n = 1, 2, \ldots$$

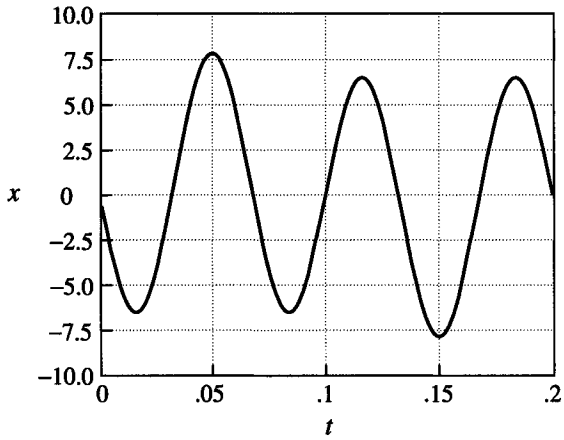


Figure 3.7 Steady state response of spring-mass system to triangle wave

Figure 3.7 shows what the response will look like if $k = 82,810$ N/m, $m = 10$ kg, and $\omega_0 = 31.4$ rad/s. For this k and m, the system's natural frequency is 91 rad/s, and thus the forcing's third harmonic ($3\omega_0 = 94.2$ rad/s) will be strongly excited owing to its proximity to the system's natural frequency. We'd therefore expect to see a large third harmonic response, which is exactly what's shown in Figure 3.7.

We've seen that the response harmonics in an input can occur near a system's natural frequency, causing a strong amplification of that harmonic. To make matters more complicated, MDOF systems and continuous systems have more than one natural frequency, any of which could potentially be excited by a harmonic of the input. Thus structural analysts must be aware of all the harmonics running around the system and how they might interact with the system's natural frequencies. If an excessive response due to resonance is found, then either the input or the system itself must be modified to move the excitation harmonic away from the natural frequency that's causing the problem.

3.3 FORCED RESPONSE VIA THE CONVOLUTION INTEGRAL

We've been concentrating on periodic excitations up until now because these are among the most common kinds of system input you'll be dealing with. There are, however, many situations in which the excitation isn't periodic and thus the approaches we've looked at aren't applicable. You know that the general solution to a forced differential equation involves both a transient and a steady state solution. So far, we've been consistently neglecting the transient motions, claiming that they'll die out as a result of dissipation, leaving us with the purely steady state response. Thus we were able to assume the existence of a steady state response (having an unknown amplitude and phase) and directly solve for this response. This section takes a different point of view. We're not going to presuppose a steady state response that will qualitatively match the excitation. Instead, we'll consider what the precise response is for *any* input and then determine the solution.

Just so you know where we're aiming, we're first going to find out how our systems respond to an impulsive force. Roughly speaking, an impulsive force is a force that occurs over a very short

span of time, like that due to a quick tap on the mass of the system. Once we've got this response, we're going to claim that *any* forcing we might run into can be approximated as a string of impulses. Thus we'll just figure out the response due to all the individual impulses and sum them together. It's similar in a way to how we used a Fourier series approach to calculate the response of the system to a periodic but nonsinusoidal input. We figured out what particular sinusoids made up the periodic signal, determined the system's response to each individually, and then summed up all the responses.

Our first question should be, What precisely is meant by an impulse? Simply (and confusingly) stated, an impulse is an input that takes place over zero time and has infinite magnitude. Clearly, such an entity can't really exist. Infinite magnitudes over zero time intervals aren't the sort of thing you can produce in the real world. This is an important point—impulses are mathematical abstractions. All we can do in the real world is approximate them. There are many ways of approximating them, and the one we'll use is probably the most easily grasped. We'll start by looking at a forcing function that's equal to $\frac{1}{\varepsilon}$ and lasts only from $t = 0$ to $t = \varepsilon$ (Figure 3.8):

$$f_\varepsilon = \frac{1}{\varepsilon}, \qquad 0 < t \leq \varepsilon$$
$$f_\varepsilon = 0, \qquad t > \varepsilon \tag{3.3.1}$$

ε is small to start, and we'll make it smaller as we go. You can see that as ε gets smaller, the magnitude of f_ε grows and the length of time it takes on nonzero values shrinks. Thus, as ε approaches zero, the magnitude goes to infinity and the input takes place over zero time. We call such a function a Dirac delta function, $\delta(t)$, and we represent it graphically by a vertical arrow, as shown in Figure 3.9. Although we can't actually attain such an input in real life, we can come close. The easiest way to approximate such an input is to take a hammer and sharply strike the mass in a system. The hammer will hit the mass and almost immediately bounce back, imparting a large force to the mass which lasts only for a short time. As we'll see later, this approach is a very common tool in vibrational analysis.

You'll note that the area under the f_ε function is always equal to 1.0, regardless of how small ε becomes. Do we have any way of knowing what such a forcing will do when applied to a dynamical

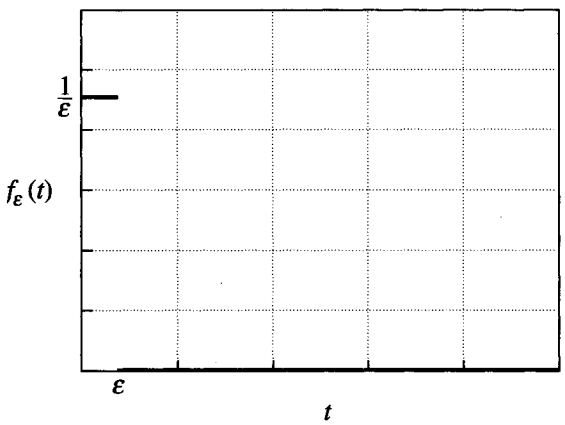


Figure 3.8 First approximation to an impulse function

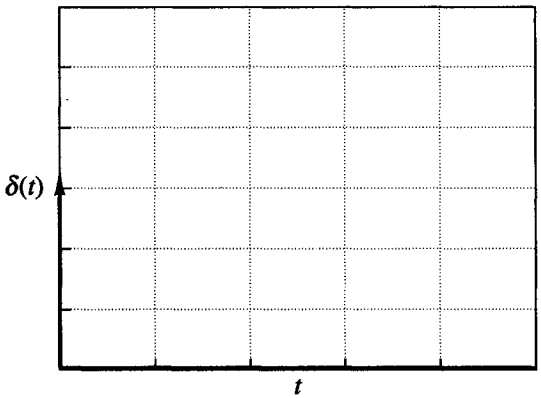


Figure 3.9 Dirac delta function

system? Actually, you've probably already learned the answer in a dynamics course. Any course in two-dimensional dynamics will cover the topic of momentum for a system and how it changes. What's taught is that the change in momentum of a mass m is equal to the *applied linear impulse*, where the applied linear impulse is defined as $\int f(t)dt$. Since this is simply the area under the f_ε curve, we see that the linear momentum must be altered as a result of the force. Thus, applying the force shown in Figure 3.8 will mean that

$$m(\dot{x}(\varepsilon) - \dot{x}(0)) = \int_0^\varepsilon \frac{1}{\varepsilon} dt = 1 \tag{3.3.2}$$

If the initial velocity is zero, this implies that the velocity at $t = \varepsilon$ is equal to

$$\dot{x}(\varepsilon) = \frac{1}{m} \tag{3.3.3}$$

As we take ε smaller and smaller (going to a limit of zero), we approach a simple change of initial conditions. As ε approaches zero, $\dot{x}(\varepsilon)$ approaches $\dot{x}(0)$. What about the displacement x, though? Is it possible that this might change also? To answer this, let's assume that it does change in response to the impulse and goes to a value x_0. We know that the velocity at time t is found from $\frac{x(t+\varepsilon)-x(t)}{\varepsilon}$. Thus, in the limit, as ε goes to zero, we'd have

$$\dot{x}(0) = \lim_{\varepsilon \to 0} \frac{x_0}{\varepsilon} \to \infty \tag{3.3.4}$$

Since we know the velocity can't go to infinity, we conclude that the displacement must actually still be zero after the impulse has been applied.

You can see the same results in a slightly different way by explicitly solving

$$m\ddot{x} = f_\varepsilon(t) \tag{3.3.5}$$

where $f_\varepsilon(t)$ is shown in Figure 3.8. Assume initial conditions $x(0) = 0$, $\dot{x}(0) = 0$ and let $m = 1$. We can integrate both sides to obtain

$$\dot{x}(t) = \frac{1}{\varepsilon}t \tag{3.3.6}$$

and

$$x(t) = \frac{1}{\varepsilon}\frac{t^2}{2} \tag{3.3.7}$$

for t between 0 and ε. (Note that the initial conditions were used in the solution to eliminate any constants of integration.) Let's assume that ε is small, say .01. Then $\dot{x}(.01) = 1.0$ and $x(.01) = .005$. It's pretty clear that the velocity is finite and the change in position is essentially zero. As ε goes to zero, this effect becomes even more pronounced, leading finally to a finite change in initial velocity and no change in displacement as ε hits zero.

What about using a Dirac delta function as the input for a more complicated system? To see, let's determine what will happen to a general spring-mass-damper system when an impulse is applied. In this case the equation of motion would be given by

$$m\ddot{x} + c\dot{x} + kx = \delta(t) \tag{3.3.8}$$

To aid in our physical understanding, let's go ahead and use the f_ε approximation for the delta function. Integrating once, from $t = 0$ to $t = \varepsilon$, gives us

$$m\left(\dot{x}(\varepsilon) - \dot{x}(0)\right) + c\left(x(\varepsilon) - x(0)\right) + k\left(\int_0^\varepsilon x\,dt\right) = 1 \tag{3.3.9}$$

Can $\int_0^\varepsilon x\,dt$ be finite (and thus part of the left-hand-side terms that add to 1.0)? If so, then $x(\varepsilon)$ must go to infinity like $\frac{1}{\varepsilon}$, since otherwise the integral would vanish. Since we already know the mass isn't going to go off to infinity in zero time, we know that the integral must therefore equal zero. Next we can ask if the displacement can be finite, letting the $c\left(x(\varepsilon) - x(0)\right)$ term contribute toward making up the 1.0 on the right-hand side. We've already seen that this would imply an infinite velocity and is therefore not possible. Thus we're left with the same result as before, namely,

$$m\left(\dot{x}(\varepsilon) - \dot{x}(0)\right) = 1 \tag{3.3.10}$$

Therefore the velocity undergoes a step change. In the limit of $\varepsilon \to 0$, this again produces the result that the effect of an applied impulsive force is to alter the initial velocity and leave the displacement undisturbed.

You can now see why using a hammer to apply an approximate Dirac delta function of force to a test object is a nice idea. If a Dirac delta is applied, the only result is to change the initial conditions of the system. Thus we'll end up with an *unforced* problem, one for which the displacement is zero and the velocity is finite. We can then observe the free response and, from it, deduce the damping, natural frequency, etc. Many techniques of modal analysis rely on just such an approach.

The other way in which the Dirac delta function is useful is in providing us with a method for determining the forced response of our system to *any* input. Figure 3.10 illustrates the first step in this approach, and Figure 3.10*a* shows the original, continuous, forcing function. The next step is to discretize the time line into segments Δ seconds long, shown in Figure 3.10*b*. If Δ is small,

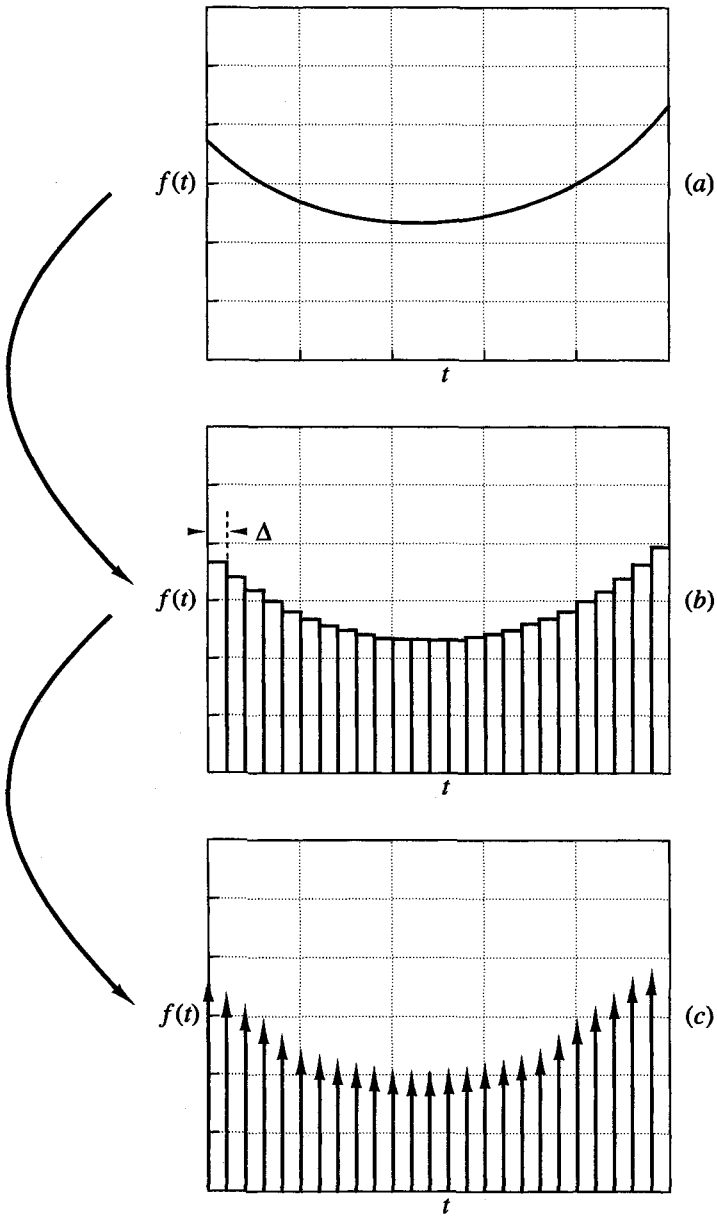


Figure 3.10 Approximation of a continuous function $f(t)$ by a train of delta functions

then the area within each four-sided box is given by the value of the function times the time interval; i.e., the first box has area $f(0)\Delta$, the second has area $f(\Delta)\Delta$, etc. The final step is to replace the function by a train of delta functions, as shown in Figure 3.10*c*. Each unit delta function is multiplied by the appropriate factor to make it match the actual function values. Thus the first delta function is multiplied by $f(0)\Delta$, the second by $f(\Delta)\Delta$, etc.

The idea here is to hit our system with this train of delta functions, rather than using the original, continuous, function. The reason for this is that we've already figured out how the system will respond to a delta function: it will have a step change in velocity and then undergo free vibration. So what we'll have is a succession of free vibration responses. The system will undergo free vibrations for Δ seconds, have a change in velocity that is due to the next impulse, and then freely vibrate for another Δ seconds. We can therefore sum up all these free vibrational responses to find out how the overall system responds.

Before doing this, let's pin down exactly the free vibration response of a spring-mass-damper system. Say our system is acted on by a Dirac delta function, with magnitude $\bar{f}$. Our equation of motion is

$$m\ddot{x} + c\dot{x} + kx = \bar{f}\delta(t) \tag{3.3.11}$$

We've seen that this is equivalent to

$$m\ddot{x} + c\dot{x} + kx = 0 \tag{3.3.12}$$

with initial conditions $x(0) = 0$ and $\dot{x}(0) = \frac{\bar{f}}{m}$.

The general solution to an unforced spring-mass-damper system is given by

$$x(t) = e^{-\zeta\omega t}\left(a_1\cos(\omega_d t) + a_2\sin(\omega_d t)\right) \tag{3.3.13}$$

where $\omega_n^2 = \frac{k}{m}$ and $\omega_d = \omega_n\sqrt{1-\zeta^2}$. Applying the $x(0) = 0$ initial condition sets a_1 to zero and applying the $\dot{x}(0) = \frac{1}{m}$ initial condition gives us $a_2 = \frac{1}{m\omega_d}$. Thus the total solution is given by

$$x(t) = \frac{1}{m\omega_d}e^{-\zeta\omega t}\sin(\omega_d t) \tag{3.3.14}$$

We'll use the symbol $h(t)$ to represent this response and remember that $h(t)$ is the *impulse response* of any spring-mass-damper system:

$$h(t) \equiv \frac{e^{-\zeta\omega_n t}}{m\omega_d}\sin(\omega_d t) \tag{3.3.15}$$

Now let's look at our original problem again. We've already approximated our input function by a string of Dirac delta functions. Therefore we know that the response from $t = 0$ to $t = \Delta$ is given by

$$x(t) = \frac{f(0)\Delta e^{-\zeta\omega_n t}}{m\omega_d}\sin(\omega_d t) = f(0)h(t)\Delta \tag{3.3.16}$$

From $t = \Delta$ to $t = 2\Delta$, the response is given by

$$x(t) = f(0)\Delta h(t) + \frac{f(\Delta)\Delta e^{-\zeta\omega_n(t-\Delta)}}{m\omega_d}\sin(\omega_d(t-\Delta))$$
$$= f(0)h(t)\Delta + f(\Delta)h(t-\Delta)\Delta \tag{3.3.17}$$

You can see the pattern shaping up here. We're going to get a longer and longer string of responses. Jumping to the response at the nth interval, we've got

$$x(t) = \sum_{i=0}^{n-1} f(i\Delta)h(t-i\Delta)\Delta \tag{3.3.18}$$

which holds from $t = (n-1)\Delta$ to $t = n\Delta$. Look at what happens if we now let the interval Δ get smaller and smaller. The summation becomes an integral, the Δ becomes a differential, and we're left with

$$x(t) = \int_0^t f(\tau)h(t-\tau)d\tau \tag{3.3.19}$$

For our problem this becomes

$$x(t) = \int_0^t f(\tau)\frac{e^{-\zeta\omega_n(t-\tau)}}{m\omega_d}\sin(\omega_d(t-\tau))d\tau \tag{3.3.20}$$

Equation (3.3.19) is called a convolution integral and, not surprisingly, this approach is called the convolution method. This may look pretty heavy-duty, but if you remember where it came from, it should seem a bit more manageable. All we're doing is trying to determine the *impulse* response at any particular point in time, and then letting this response remain. It's even easier to see what's going on if we use a particular time t_1 rather than t. In this case the response at t_1 is given by

$$x(t_1) = \int_0^{t_1} f(\tau)h(t_1-\tau)d\tau \tag{3.3.21}$$

If we think about it in this way we see that the response at t_1 depends upon $f(\tau)$, with $f(\tau)$ going from $f(0)$ to $f(t_1)$ and upon $h(t_1-\tau)$, with $h(t_1-\tau)$ going from $h(t_1)$ to $h(0)$. For instance, take the first part of the integral, near $\tau = 0$. This is approximately equal to $f(0)h(t_1)\Delta$, where we're just taking a Δ-wide slice of time. This is just the response due to an initial impulse. Since we're looking at the response at time t_1, we let the system continue to vibrate for that long (i.e., we use $h(t_1)$).

Now look at the end of the integral. In this case τ is almost equal to t_1, say $t_1 - \Delta$, and the chunk of the integral near the end is equal to $f(t_1-\Delta)h(\Delta)\Delta$. This indicates that some of the response is due to the latest excitation. This excitation occurred at $t_1 - \Delta$ and is given by $f(t_1-\Delta)\Delta$. The free vibration lasts for only Δ seconds in this case; thus we use $h(\Delta)$.

If you look at the discrete way we first approached the integral, and think about what we've just gone over, you'll see the same result is indicated by both approaches. Taken in total, this integral operation is saying to take the response due to the early forcing and remember it. Then add to this the

response due to the force that occurred a little bit later. Then add the part due to even later forcings. Continue doing this until you're at the time you care about.

Although (3.3.19) is totally valid, you may find it easier to express it in a slightly different way. To derive the new form, we'll define a new variable γ, where we'll say that $\gamma = t - \tau$. Remember that t is fixed in the convolution. Therefore if we take a differential of γ we'll get

$$d\gamma = -d\tau \tag{3.3.22}$$

Eliminating τ in favor of γ in (3.3.19) gives us

$$x(t) = -\int_t^0 f(t-\gamma)h(\gamma)d\gamma \tag{3.3.23}$$

We found the new limits of integration by substituting what τ is equal to at these limits (0 and t) into our expression for γ. We can flip the limits of integration (which simply generates a -1 factor), cancel this -1 with the existing -1 in front of the integral, and get

$$x(t) = \int_0^t f(t-\gamma)h(\gamma)d\gamma \tag{3.3.24}$$

Since γ is just a dummy variable (it doesn't show up in the final result), we can rename it τ so that the equation looks more like (3.3.19):

$$x(t) = \int_0^t f(t-\tau)h(\tau)d\tau \tag{3.3.25}$$

This is identical to (3.3.19) except that we're putting the time shift into the forcing function f, rather than into the impulse response h. You should use whichever form is more convenient for your particular problem.

What we've now derived is a general way of determining the response of a system to any input *if* you've already found out the response of the system to a delta input. Let's work out an example to pin it down a little more.

Example 3.2

Problem Consider the damped system and excitation shown in Figure 3.11. The excitation is a relatively simple one, namely, a step input. This is the sort of situation that would occur if you could have a mass suspended from a spring in the absence of gravity and, at $t = t_0$, suddenly switch gravity on. For clarity, we'll let $t_0 = 0$. Letting it be any other value doesn't affect the solution at all except to shift the time that the mass starts to move from $t = 0$ to $t = t_0$. Let's find the response to this input.

Solution To solve this problem by the method outlined in the preceding section, we'll need the response of the system to a unit delta function, i.e., the solution to

$$m\ddot{x} + c\dot{x} + kx = \delta(t)$$

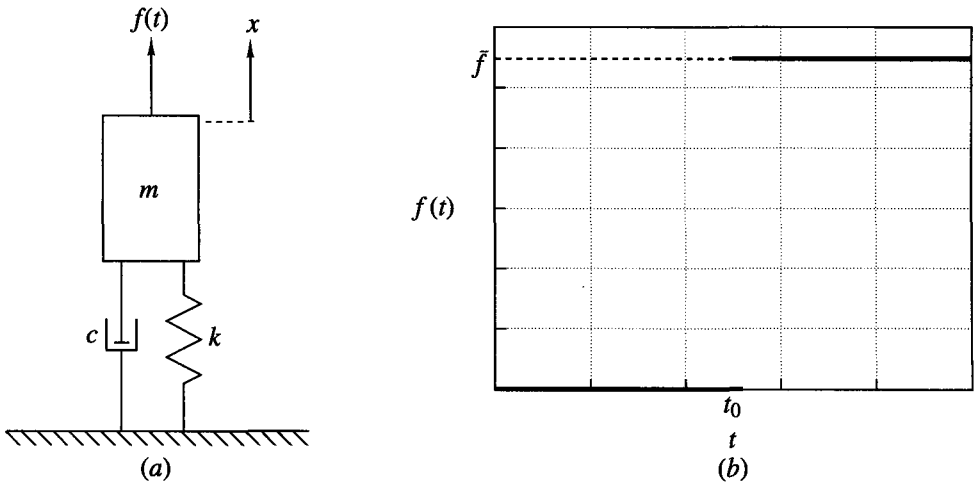


Figure 3.11 Damped system subjected to a step input force

We already know from (3.3.15) that the solution to this problem is given by

$$h(t) = \frac{e^{-\zeta\omega_n t}}{m\omega_d}\sin(\omega_d t)$$

Knowing $h(t)$, we can go ahead and use the convolution integral (3.3.25) to find the response to the illustrated step input

$$x(t) = \int_0^t h(\tau)f(t-\tau)d\tau = \frac{\bar{f}}{m\omega_d}\int_0^t e^{-\zeta\omega_n\tau}\sin(\omega_d\tau)\,d\tau \tag{3.3.26}$$

At this point we could simply use a table of integrals to evaluate (3.3.26). What we'll do instead is show how to calculate the answer. That way you'll be able to carry on even if you don't have an integral table handy.

If you try integration by parts, you'll still end up with an integral involving the product of a sine (or cosine) and an exponential. The trick to getting the answer is to integrate by parts *twice*. In this way we'll end up with an equation having the integral we want to evaluate on the left and also on the right. By combining the two, we'll be able to solve the problem. It goes like this. For the first integration by parts ($\int u\,dv = uv - \int v\,du$) we identify dv as $\sin(\omega_d\tau)d\tau$ and u as $e^{-\zeta\omega_n\tau}$. This gives us

$$\frac{\bar{f}}{m\omega_d}\int_0^t e^{-\zeta\omega_n\tau}\sin(\omega_d\tau)\,d\tau$$
$$= \frac{\bar{f}}{m\omega_d^2}\left(1 - e^{-\zeta\omega_n t}\cos(\omega_d t) - \zeta\omega_n\int_0^t e^{-\zeta\omega_n\tau}\cos(\omega_d\tau)d\tau\right)$$

When integrating the $e^{-\zeta\omega_n\tau}\cos(\omega_d\tau)d\tau$ term, we'll use $u = e^{-\zeta\omega_n\tau}$ and $dv = \cos(\omega_d\tau)$. This will give us

$$\begin{aligned}
&\frac{\bar{f}}{m\omega_d}\int_0^t e^{-\zeta\omega\tau}\sin\left(\omega_d(\tau)\right)d\tau \\
&= \frac{\bar{f}}{m\omega_d^2}\left(1 - e^{-\zeta\omega_n t}\cos(\omega_d t)\right) - \frac{\zeta\omega_n\bar{f}}{m\omega_d^3}e^{-\zeta\omega_n t}\sin(\omega_d t) \\
&\quad - \frac{\bar{f}(\zeta\omega_n)^2}{m\omega_d^3}\int_0^t e^{-\zeta\omega_n\tau}\sin(\omega_d\tau)d\tau
\end{aligned}$$

You can see that we've got the same integrals on the left and on the right. Combining them on the left-hand side of the equation gives us

$$\begin{aligned}
&\frac{\bar{f}}{m\omega_d(1-\zeta^2)}\int_0^t e^{-\zeta\omega_n\tau}\sin\left(\omega_d\tau\right)d\tau \\
&= \frac{\bar{f}}{k(1-\zeta^2)}\left(1 - e^{-\zeta\omega_n t}\cos(\omega_d t)\right) - \frac{\zeta\bar{f}}{k(1-\zeta^2)^{\frac{3}{2}}}e^{-\zeta\omega_n t}\sin(\omega_d t)
\end{aligned}$$

Some of the ω_d's were reexpressed in terms of ω_n and ζ, and $m\omega_n^2$ was re-expressed as k in the preceding expression. Multiplying by $(1-\zeta^2)$ gives us our final expression for $x(t)$

$$x(t) = \frac{\bar{f}}{k}\left(1 - e^{-\zeta\omega_n t}\cos(\omega_d t)\right) - \frac{\zeta\bar{f}}{k\sqrt{1-\zeta^2}}e^{-\zeta\omega_n t}\sin(\omega_d t)$$

This solution is illustrated in Figure 3.12. Note that the response starts at zero, grows to a maximum, and then dies down as it oscillates about $\frac{\bar{f}}{k}$. Of course, this steady state response was predictable from the original differential equation by setting all the time-varying parts ($\ddot{x}$ and $\dot{x}$) to zero and solving the resulting equation. Varying the damping will serve to reduce or increase the rate of decay of the oscillations.

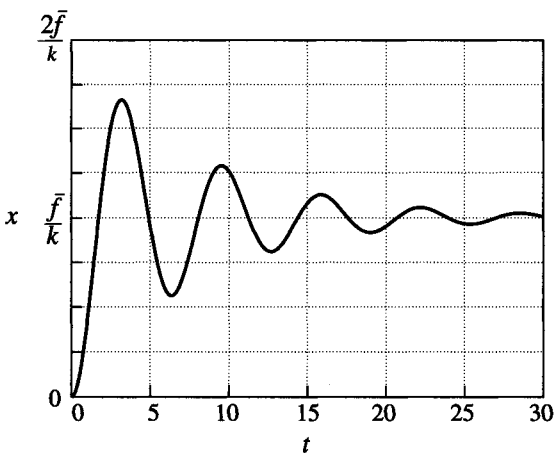


Figure 3.12 Displacement response for a damped system subjected to a step force input

Example 3.3

Problem Our example will be interesting for two reasons. The first is that we'll see what the response looks like for a finite ramp input force. All right, that really isn't all that interesting, is it? But the second reason is better. After we've found the response to a finite ramp, we'll ensure that the area within the ramp is always equal to 1.0 and we'll reduce the duration of the ramp to zero. This means that the ramp will look a lot like the Dirac delta function, with an infinite value, zero duration, and unit area. The interesting question will be whether the system responds to this input as it did to the force profile shown earlier in Figure 3.8.

The force profile is shown in Figure 3.13*b*. As you can see, the force ramps up linearly from zero to a maximum of $\bar{f}$ (at $t = t_0$). Our equation of motion is given by

$$m\ddot{x} + kx = f(t)$$

or

$$\ddot{x} + \omega_n^2 x = \frac{f(t)}{m}$$

We can find $h(t)$ either by solving the free vibration problem subjected to a unit velocity or, more simply, just using (3.3.15) and setting $\omega_d = \omega_n$ and $\zeta = 0$. This results in

$$h(t) = \frac{1}{m\omega_n} \sin(\omega_n t)$$

Using $h(t)$ in our convolution integral gives us

$$x(t) = \frac{1}{m\omega_n} \int_0^t f(t-\tau) \sin(\omega_n \tau) d\tau \tag{3.3.27}$$

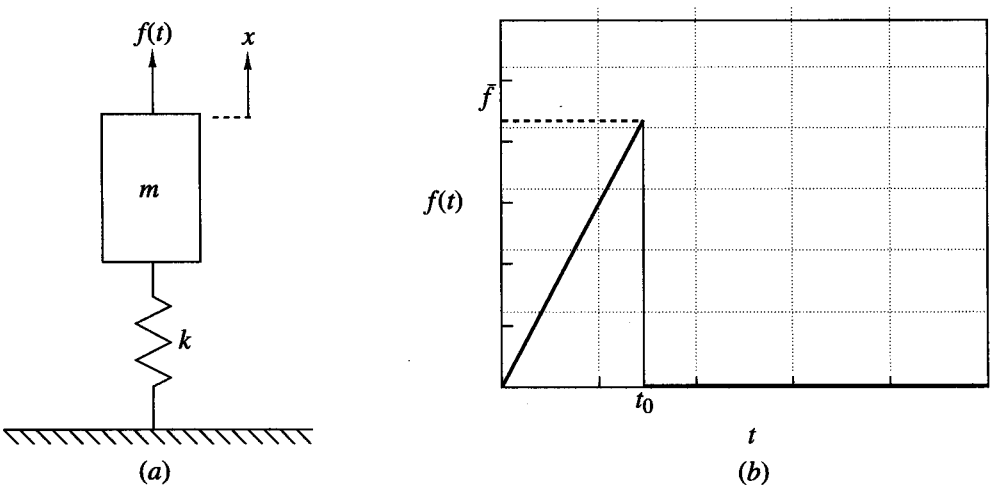


Figure 3.13 Undamped system subjected to a ramp input force

If t is less than t_0, then we can simply use this expression for $x(t)$. Since the forcing function is zero for $t > t_0$, we'd have to solve for $x(t_0)$ and $\dot{x}(t_0)$ and then solve for the system's free vibrational response for all $t > t_0$, using $x(t_0)$ and $\dot{x}(t_0)$ as initial conditions.

To solve (3.3.27), we'll need to reexpress $f(t)$ more explicitly. Since $f(t)$ is linearly increasing from 0 to $\bar{f}$, it's given by

$$f(t) = \frac{\bar{f}}{t_0}t, \qquad 0 < t \leq t_0$$

Using this form for $f(t)$, we find that $x(t)$ becomes

$$x(t) = \frac{\bar{f}}{m\omega_n t_0}\int_o^t \sin(\omega_n \tau)(t-\tau)d\tau$$

Expanding this yields

$$x(t) = \frac{\bar{f}}{m\omega_n t_0}\left(t\int_o^t \sin(\omega_n \tau)d\tau - \int_o^t \tau\sin(\omega_n \tau)d\tau\right)$$

We were able to pull the t out of the first integral because it is considered to be a constant during the integration with respect to τ. Integrating these (using integration by parts for the second integral) yields

$$x(t) = \frac{\bar{f}}{m\omega_n^2 t_0}\left(t - \frac{\sin(\omega_n t)}{\omega_n}\right)$$

If we cared to, we could now use this formula to find $x(t_0)$ and $\dot{x}(t_0)$ and then apply these as initial conditions to determine the free vibrations of the subsequent motion. Instead, what we'll do is let $\bar{f} = \frac{2}{t_0}$. For this value of $\bar{f}$ you can see that the area under the $f(t)$ curve is equal to 1.0. The solution for $x(t)$ is now given by

$$x(t) = \frac{2}{m\omega_n^2 t_0^2}\left(t - \frac{\sin(\omega_n t)}{\omega_n}\right) \tag{3.3.28}$$

The next step is to ask what $x(t_0)$ and $\dot{x}(t_0)$ are as t_0 goes to zero. For t_0 small, (3.3.28) becomes

$$x(t_0) \approx \frac{2}{m\omega_n^2 t_0} - \frac{2}{m\omega_n^3}\frac{\omega_n t_0}{t_0^2} = 0$$

Differentiating (3.3.28) gives us

$$\dot{x}(t) = \frac{2}{m\omega_n^2 t_0^2}(1 - \cos(\omega_n t))$$

Letting $t = t_0$ and expanding out $\cos(\omega_n t)$ to include the first two terms gives us

$$\dot{x}(t_0) = \frac{2}{m\omega_n^2 t_0^2}\left(1 - \left(1 - \frac{(\omega_n t_0)^2}{2}\right)\right)$$

which leads to

$$\dot{x}(t_0) = \frac{2}{m\omega_n^2 t_0^2}\frac{\omega_n^2 t_0^2}{2} = \frac{1}{m}$$

There we have it! We've found that as t_0 goes to zero, the application of the ramp input has served to alter the system initial conditions from $x(0) = 0$ and $\dot{x}(0) = 0$ to $x(0) = 0$ and $\dot{x}(0) = \frac{1}{m}$, exactly what happened when we applied the delta function that came from f_ε. Thus a delta function that's found from letting the duration of a step input go to zero, while holding the area fixed at 1.0 has the same effect as a delta function formed from a ramp input in which the duration of the ramp goes to zero and the integral is held at 1.0. As you now might guess, just about any function you can come up with that has a vanishingly small duration and an area equal to 1.0 will act in the same way. This fact is the reason for much of the success in impact testing in modal analyses. For these tests the structure under consideration must be struck with an instrumented hammer. The hope is that the force input of the hammer acts like a Dirac delta function. Sophisticated software can then capture the output of the sensors attached to the structure and determine the system properties. Obviously the hammer hit can't really input a Dirac delta function. Our results here show that this doesn't matter. As long as the hammer strike is quick, it is well approximated by a Dirac delta function.

Before leaving this section, we'll discuss a couple of additional properties of the Dirac delta function because they'll be useful to know later on. We've already seen that the Dirac delta function is defined as a function that's zero everywhere except at zero (where it's equal to infinity). The integral over the delta function is always equal to 1.0. Thus

$$\int_{-\infty}^{\infty} \delta(t)dt = 1 \tag{3.3.29}$$

We can be a little more general and say that the delta occurs not at $t = 0$ but at any time t_1. Such a function is shown in Figure 3.14. Since all we've done is displace the nonzero part of the delta function, the integral over t will still be equal to 1.0.

A helpful property associated with delta functions is that

$$\int_{-\infty}^{\infty} f(t)\delta(t - t_1)dt = f(t_1) \tag{3.3.30}$$

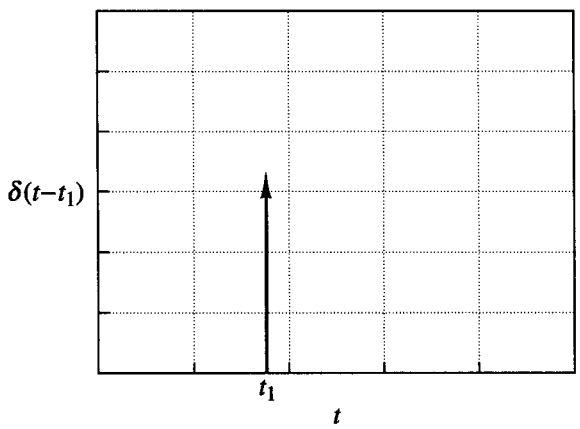


Figure 3.14 Delta function displaced in time

If we recall that the delta function can be viewed as the limiting case of the f_ε function shown in Figure 3.8 (and letting the nonzero part occur at $t = t_1$), this property is pretty easy to see. Since the delta function is zero almost everywhere, most of the integral in (3.3.30) is just equal to zero. The only nonzero contribution occurs right where the delta becomes nonzero. If ε is small, $f(t)$ is approximately equal to $f(t_1)$ over the entire interval. Thus the integral is approximated by

$$\int_{-\infty}^{\infty} f(t)\delta(t - t_1)dt \approx f(t_1)\frac{1}{\varepsilon}\varepsilon \tag{3.3.31}$$

In the limit of $\varepsilon \to 0$, this becomes an equality. Thus we see that the ε's cancel out, leaving us with

$$\int_{-\infty}^{\infty} f(t)\delta(t - t_1)dt = f(t_1) \tag{3.3.32}$$

Example 3.4

Problem Express the following force input as a Fourier series expansion:

$$f(t) = \sum_{n=-\infty}^{\infty} \delta(t - n)$$

Solution

$$f(t) = \sum_{m=0}^{\infty} a_m \cos(m\omega_0 t) + \sum_{m=1}^{\infty} b_m \sin(m\omega_0 t)$$

$$\omega_0 = \frac{2\pi}{1} = 2\pi$$

The function in Figure 3.15 is even and thus $b_m = 0$. The a_i's are given by

$$a_0 = \int_0^1 f(t)dt, \quad a_m = 2\int_0^1 f(t)\cos(2\pi m t)dt$$

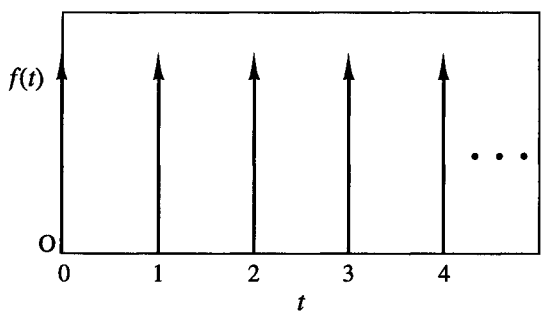


Figure 3.15 Function for Example 3.4

For simplicity, shift the integration interval to $-\frac{1}{2}$ to $\frac{1}{2}$. In this case we'll have

$$a_0 = \int_{-\frac{1}{2}}^{\frac{1}{2}} \delta(t)dt = 1$$

$$a_m = 2\int_{-\frac{1}{2}}^{\frac{1}{2}} \delta(t)\cos(2\pi mt)dt = 2\cos(0) = 2$$

$$f(t) = 1 + 2\sum_{m=1}^{\infty} \cos(2\pi mt)$$

3.4 SHOCK RESPONSE

A special class of transient loadings are those involving shock. As mentioned earlier, shock loadings can come about in a variety of ways. Explosions always generate shocks. When an object is suddenly brought to rest after falling, the result is a shock. When a car hits a pothole, a shock is produced that then affects the car and anything in it. Obviously, when one has a sharp loading of this sort, concern over an object's structural integrity becomes acute. The acceleration levels produced by shock loading are high and can easily lead to breakage within the system.

Some common shock profiles are shown in Figure 3.16*a*–*c*. In Figure 3.16*a*, the force input rises quickly to some maximum value and then stays constant at that value. This might be the case for the loading within a hydraulic system. Once the system is pressurized, the force on an interior surface will be higher than it was before pressurization. If the rise in pressure is rapid, the result is a shock load. In this case the shock is caused by the rapidly increase in force, not by the steady state force that exists after t_1.

Figure 3.16*b* shows a shock load that consists of a constant force with a short duration t_0. This sort of shock can occur as a result of a blast wave, for which the object in the path of the blast will feel a high level of force for a short time.

Finally, Figure 3.16*c* shows a common approximation to shock loadings, called a half-sine loading. Just as you'd expect from the name, the load has the profile of half a sine wave and then remains at zero for later times. If t_1 is small and the amplitude of the half-sine is large, the result is a shock load. Engineering experience and insight will often be needed to decide whether the loading shown in Figure 3.16*b* or *c* is the more appropriate for the actual shock being considered. If the time duration of the shock is short enough, it might be sufficient to simply view the input as a Dirac delta function.

Once we know the profile of the shock load, it's straightforward to apply the convolution approach to determine the response. Alternatively, considering the speed of modern computers, we might simply numerically integrate the equations of motion, using the shock profile as the input excitation. In either case, once we've determined the response, we can then characterize the effect of the shock. Commonly, we're concerned with the *peak* acceleration levels that the shock will induce

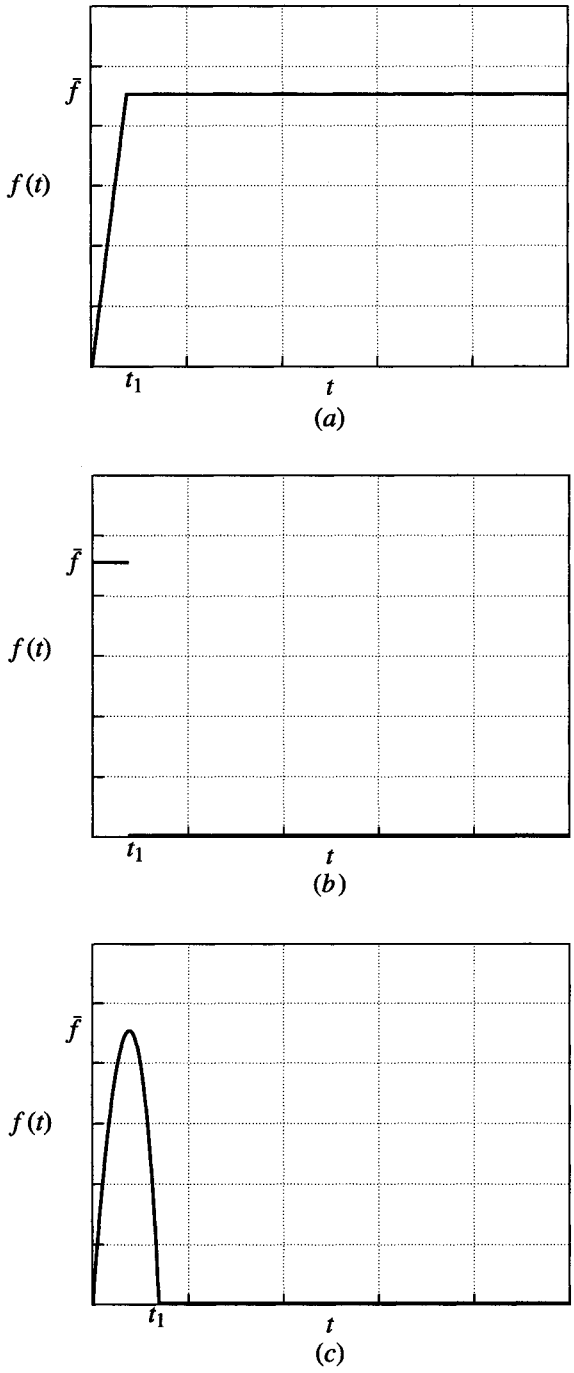


Figure 3.16 (*a*) Abrupt change in force level, (*b*) impulsive force, and (*c*) half-sine input force

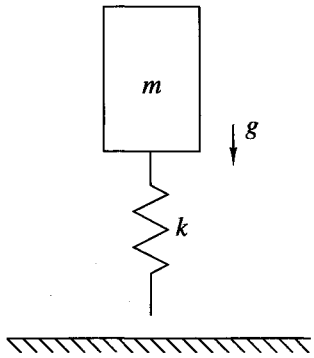


Figure 3.17 Spring-mass system subjected to shock due to impact

in the system. Thus we can plot peak acceleration versus some shock parameter (say shock duration) and get a feel for how the system will be affected by different shocks.

Let's consider the case for which a test object has been dropped and contacts the ground. We'll determine what the acceleration levels are for the mass and see how they vary as a function of contact velocity. Finally, we'll relate contact velocity to the drop height. This will allow us to set safe limits with respect to how high the structure should be carried during movement, since the height at which the object is dropped is directly related to the impact velocity. The system to be examined, shown in Figure 3.17, consists of a spring and a mass. When dropped, it will accelerate owing to gravity and eventually contact the floor with a velocity v. At the instant it strikes the floor, the spring will be uncompressed. We'll measure x from the inital position of the mass upon just contacting the floor. We'll also orient x downward so that the initial velocity upon contact is positive. The relevant equation of motion is given by

$$m\ddot{x} + kx = mg \tag{3.4.1}$$

with initial conditions $x(0) = 0$ and $\dot{x}(0) = v$.

Dividing by m gives us

$$\ddot{x} + \omega_n^2 x = g \tag{3.4.2}$$

The general solution for this problem is given by

$$x(t) = a_1 \cos(\omega_n t) + a_2 \sin(\omega_n t) + \frac{g}{\omega_n^2} \tag{3.4.3}$$

where the sine and cosine are the homogeneous solution and $\frac{g}{\omega_n^2}$ is the particular solution, i.e., the one due to the gravity loading. This problem therefore falls into the category of Figure 3.16*a*, one in which a constant loading is suddenly applied to the system.

Applying the initial condition $x(0) = 0$ tells us that $a_1 = -\frac{g}{\omega_n^2}$, and subsequent application of $\dot{x}(0) = v$ gives us $a_2 = \frac{v}{\omega_n}$. Thus our complete solution is given by

$$x(t) = \frac{g}{\omega_n^2}(1 - \cos(\omega_n t)) + \frac{v}{\omega_n}\sin(\omega_n t) \tag{3.4.4}$$

To find the accelerative response, we need to differentiate twice, yielding

$$\ddot{x}(t) = g\cos(\omega_n t) - v\omega_n \sin(\omega_n t) \tag{3.4.5}$$

We know from our work with harmonic vibrations that this response can be expressed as $a\cos(\omega_n t + \phi)$, where a is given by

$$a = \sqrt{g^2 + (v\omega_n)^2} \tag{3.4.6}$$

Therefore the maximum acceleration magnitude is given by a.

It's always nice to normalize our plots, and a convenient way to normalize the acceleration level is to divide by gravity. Thus our normalized maximum acceleration is given by $\frac{a}{g}$ or, expressed slightly differently,

$$\left|\frac{\ddot{x}}{g}\right|_{\max} = \sqrt{1 + \left(\frac{v\omega_n}{g}\right)^2} \tag{3.4.7}$$

An equation you've undoubtedly seen in your dynamics course is

$$\ddot{x}dx = \dot{x}d(\dot{x}) \tag{3.4.8}$$

a result that allows us to immediately determine the relationship between the drop height x_h and the impact velocity v. Using a constant acceleration (equal to g) to integrate this equation gives us

$$gx_h = \frac{v^2}{2} \tag{3.4.9}$$

Solving for v gives us

$$v = \sqrt{2gx_h} \tag{3.4.10}$$

Thus our normalized maximum acceleration versus drop height becomes

$$\left|\frac{\ddot{x}}{g}\right|_{\max} = \sqrt{1 + \frac{2x_h\omega_n^2}{g}} \tag{3.4.11}$$

Finally, look again at (3.4.2). As we've discussed, the steady state deflection of the mass due to gravity is found by setting the $\ddot{x}$ term to zero, giving us

$$\omega_n^2 x_{\text{eq}} = g \tag{3.4.12}$$

or

$$x_{\text{eq}} = \frac{g}{\omega_n^2} \tag{3.4.13}$$

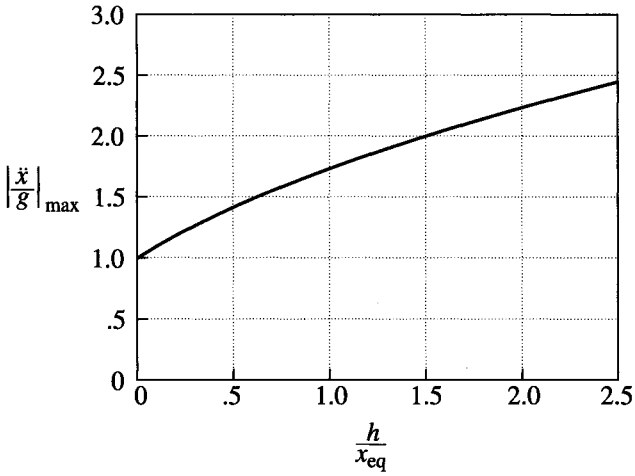


Figure 3.18 Normalized maximum acceleration versus normalized distance fallen

Interestingly, you'll see that this term pops up in (3.4.11). Substituting x_{eq} for $\frac{g}{\omega_n^2}$ in (3.4.11) gives us our final expression:

$$\left|\frac{\ddot{x}}{g}\right|_{max} = \sqrt{1 + \frac{2x_h}{x_{eq}}} \tag{3.4.14}$$

Figure 3.18 shows a plot of this maximal acceleration versus the normalized drop height $\frac{x_h}{x_{eq}}$. You can see that the maximal acceleration increases as we move along the horizontal axis. This certainly makes sense; as the item is dropped from greater heights, the maximal acceleration levels should increase. We can get some more insight by recalling that $\frac{x_h}{x_{eq}}$ is the same as $\frac{1}{2}(\frac{v\omega_n}{g})^2$. Since increases in v move us along the horizontal axis, we see that increasing impact velocities increase our maximal accelerations. This fact is directly related to the preceding observation. A new observation, however, is that increasing ω_n also moves us to the right on the horizontal axis, thus increasing maximal acceleration levels. This gives us some useful design insight. We now know that increasing the natural frequency of a system is bad, at least as far as the impact loads are concerned if the system is dropped. We might want a higher natural frequency for other reasons (such as strength), but this tells us that a trade-off we'll then have to live with is an increased susceptibility to certain shock loadings. Often one will find that structures that deal with shock very well are not particularly good with regard to steady state excitation and vice versa. Once again this is a classic case of how one must carefully consider the different factors of a design, weigh their importance, and then come up with the best overall design that produces acceptable performance in all the relevant categories.

Example 3.5

Problem Determine how one can drop a mass of 1 kg without the maximal acceleration levels exceeding $10g$. Assume that the floor is massless and has an effective stiffness coefficient of 10,000 N/m. $g = 9.81$ m/s^2.

Solution From (3.4.11) we have

$$|\ddot{x}|_{\max} = g\sqrt{1 + \frac{2x_h\omega_n^2}{g}}$$

ω_n for this example is given by $\sqrt{\frac{10{,}000}{10}} = 31.6$ m/s^2 Thus we have to solve

$$10g = g\sqrt{1 + \frac{2x_h\omega_n^2}{g}}$$

for x_h. Rearranging this gives us

$$x_h = \frac{99g}{2\omega_n^2} = .486 \text{ m}$$

As a further illustration of how we can examine shock responses, we'll again look at a falling vibratory system, but this time we'll include damping. Figure 3.19 shows the system configuration. In this case, the analysis is a bit more complex, since we'll have to account for the exponential decay of the vibrations.

In this case we won't be able to simply find the overall amplitude of the sinusoidally varying acceleration and say it is equal to the maximum acceleration, as we did in Example 3.5. For this system we'll have to actually find the time at which the maximal response occurs and then evaluate its magnitude. Figure 3.20 shows a typical amplitude and acceleration response ($\zeta = .2$, $\omega_n = 1$ rad/s, $v = 10$ m/s) for such a system, while Figure 3.21 shows how the acceleration profile changes if the natural frequency of the system is increased to 100 rad/s. Clearly, increasing the natural frequency has increased the overall magnitudes of the accelerations.

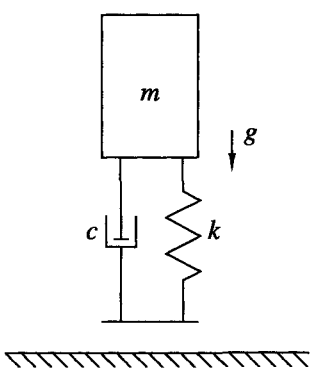


Figure 3.19 Spring-mass damper subjected to shock due to impact

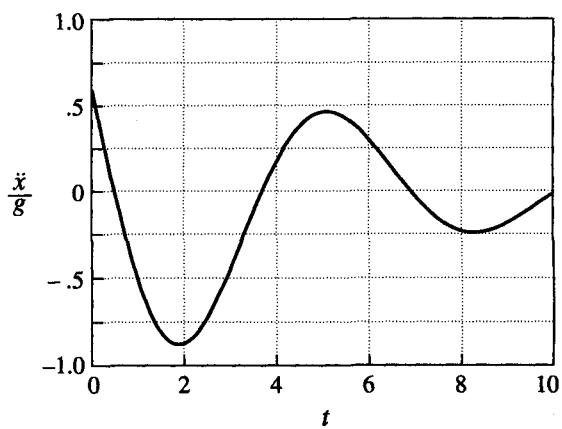


Figure 3.20 Normalized acceleration for a damped system impacting a rigid floor

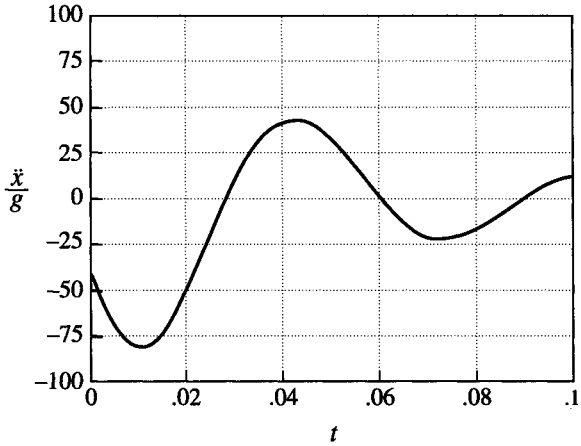


Figure 3.21 Normalized acceleration for a damped system impacting a rigid floor ($\omega_n = 100$ rad/s)

The equation of motion is given by

$$m\ddot{x} + c\dot{x} + kx = mg \tag{3.4.15}$$

or

$$\ddot{x} + 2\zeta\omega_n\dot{x} + \omega_n^2 x = g \tag{3.4.16}$$

with $x(0) = 0$ and $\dot{x}(0) = v$. The general solution for this problem is given by

$$x(t) = \frac{g}{\omega_n^2} + e^{-\zeta\omega_n t}(a_1\cos(\omega_d t) + a_2\sin(\omega_d t)) \tag{3.4.17}$$

where ω_d is the damped frequency of oscillation. Applying the initial conditions gives us

$$a_1 = -\frac{g}{\omega_n^2} \tag{3.4.18}$$

and

$$a_2 = \frac{v - \frac{\zeta g}{\omega_n}}{\omega_d} \tag{3.4.19}$$

Differentiating twice gives us the acceleration:

$$\ddot{x}(t) = e^{-\zeta\omega_n t}\left((g - 2\zeta\omega_n v)\cos(\omega_d t) + \left(\frac{v\omega_n(2\zeta^2 - 1) - g\zeta}{\sqrt{1-\zeta^2}}\right)\sin(\omega_d t)\right) \tag{3.4.20}$$

As you can see, this is quite a bit more complicated than the undamped case was. If we were just going to check numerically for when the absolute value of the acceleration was a maximum, we wouldn't need to go any further. If we want to determine the peak response analytically, however, we'll have to differentiate once again. Doing so, setting the result to zero (to find the maximum value), and

rearranging slightly will give us

$$\tan(\omega_d t_{\max}) = \frac{\omega_d b_2 - \omega_n \zeta b_1}{\omega_d b_1 + \omega_n \zeta b_2} \tag{3.4.21}$$

where $b_1 = g - 2\zeta\omega_n v$ and $b_2 = \frac{v\omega_n(2\zeta^2-1)-g\zeta}{\sqrt{1-\zeta^2}}$

To solve for $t_{\max}$, we've got to take the inverse tangent

$$\omega_d t_{\max} = \tan^{-1}\left(\frac{\omega_d b_2 - \omega_n \zeta b_1}{\omega_d b_1 + \omega_n \zeta b_2}\right) \tag{3.4.22}$$

This will then allow us to solve for the time at which the maximum response occurs.

Unfortunately, the nice way we plotted normalized acceleration versus normalized drop height won't really work here. The variables are too complex to allow a simple normalization such as we used before. It's more convenient in this case to plot the acceleration-to-gravity ratio versus the impact velocity. We'd then need to make a secondary calculation to relate impact velocity to drop height.

The results of following through on the procedure just outlined are shown in Figure 3.22. The normalized, maximum acceleration is plotted versus impact velocity for ζ ranging from 0 to .4. The natural frequency was taken to be 1.0 rad/s for this plot. There are a few observations that can be made from these results. First, you'll note that the maximal acceleration ratio is sometimes less than 1. This might seem odd, since the object was in a $1g$ free-fall before impact, but what we're looking at is the maximum acceleration ratio after impact has occurred. Once the object has struck the ground, the acceleration is simply a function of the initial velocity and the damping and stiffness characteristics. As you might guess, impacts typically result in accelerative loadings much higher than g, and thus this observation is mainly of academic interest.

The second observation is that the acceleration ratio is growing linearly for large values of impact velocity. You can see why this happens from looking at (3.4.20). As v increases, it will come to dominate the coefficients of both the cosine and sine terms. Since v appears linearly, we'd expect

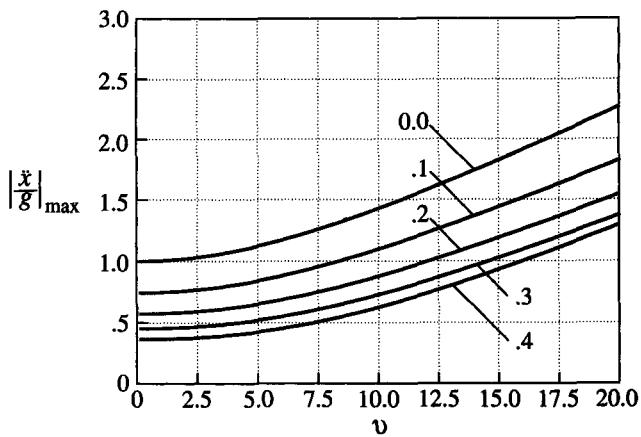


Figure 3.22 Normalized maximum acceleration versus impact velocity for damped system (damping factors indicated on graph)

a linear increase in our plot. This fact simply tells us that for large values of v, as the impact velocity increases, so does the maximal acceleration.

Finally, it's clear for this range of parameters that increasing the damping serves to reduce the maximal acceleration levels. Thus damping is another useful design tool to use in deciding how to build systems to withstand shock loading.

3.5 HOMEWORK PROBLEMS

Section 3.2

3.1. Find the response of the system illustrated in Figure P3.1 to the input force shown.

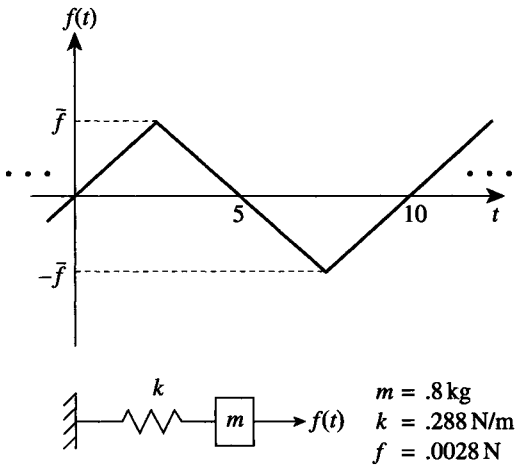


Figure P3.1

3.2. How many terms have to considered in a Fourier expansion of the signal shown in Figure P3.2 before the individual harmonic components are less than 5% of the magnitude of the fundamental component (the one at ω_0)?

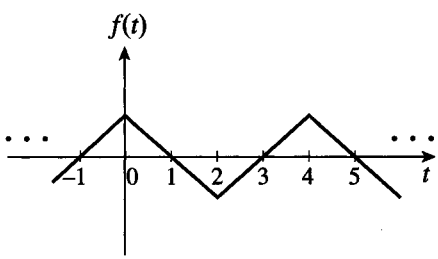


Figure P3.2

3.3. Calculate and plot several Fourier approximations (numerically is fine) for the function shown in Figure P3.3. How many terms are needed before the fit looks reasonable?

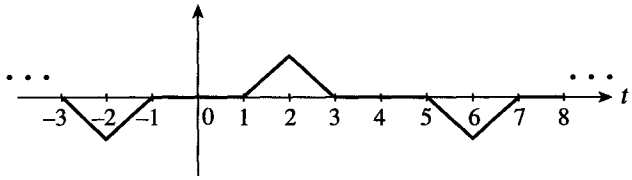


Figure P3.3

3.4. Plot the response of the system illustrated in Figure P3.4 to the illustrated seismic input.

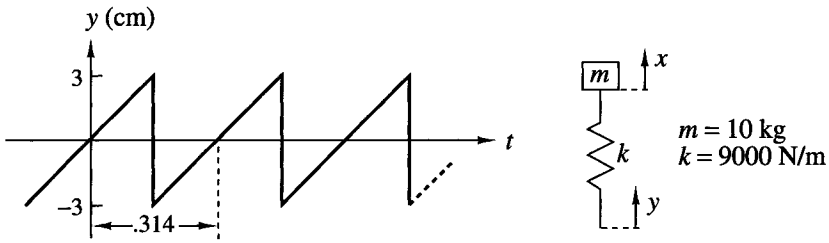
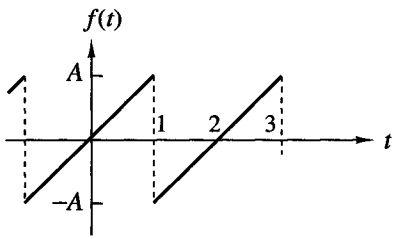


Figure P3.4

3.5. Given the Fourier series solution for $f(t)$ in Figure P3.5, construct the solution for $g(t)$ also shown in terms of the a_i and b_i



$$f(t) = \sum_{i=1}^{\infty} a_i \sin(n\omega_0 t) + \sum_{i=1}^{\infty} b_i \cos(n\omega_0 t)$$

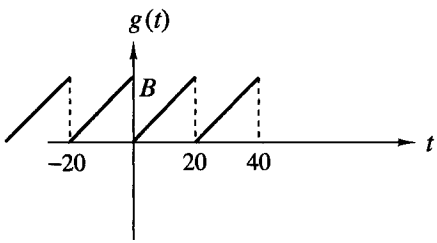


Figure P3.5

3.6. How do the coefficients of the Fourier representation change if, instead of the signal $f(t)$, shown in Figure P3.5, your signal looks like that in Figure P3.6?

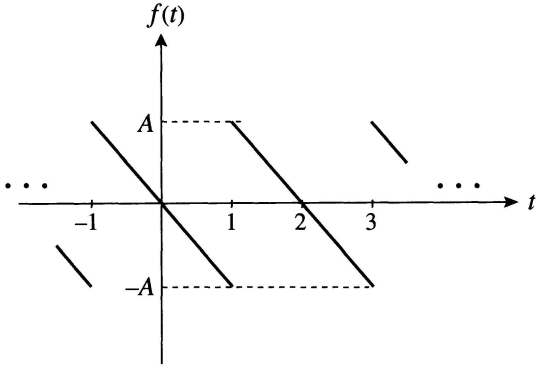


Figure P3.6

3.7. How do the coefficients of the Fourier representation change if, instead of the signal g(t), shown in Figure P3.5, your signal looks like that in Figure P3.7?

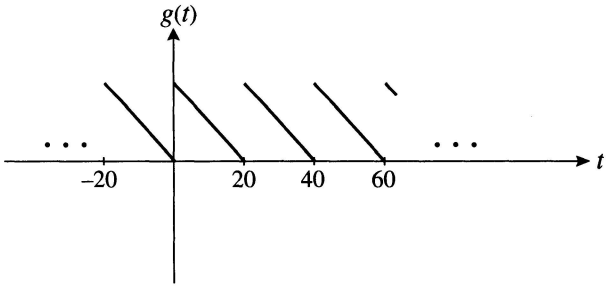


Figure P3.7

3.8. For the system shown in Figure P3.8, $m = 20$ kg and $k = 400$ N/m. The base of the system is given a vertical displacement $y(t)$ in the form of a sawtooth function with amplitude $A = 3$ cm and period $T = 2$ seconds. Compute the response of the system and identify the dominant response harmonic.

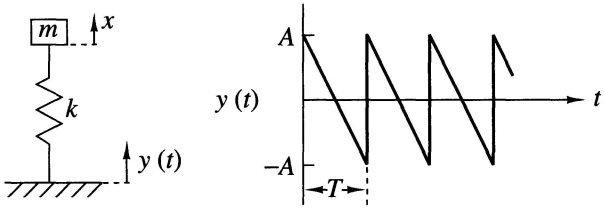


Figure P3.8

3.9. Compute and plot the response for the system illustrated in Figure P3.9 for the input force also shown. You'll note that the second harmonic ($2\omega_0 = 1.57$) is very close to the system's natural frequency. Since the damping is equal to zero, why isn't this harmonic dominating the response?

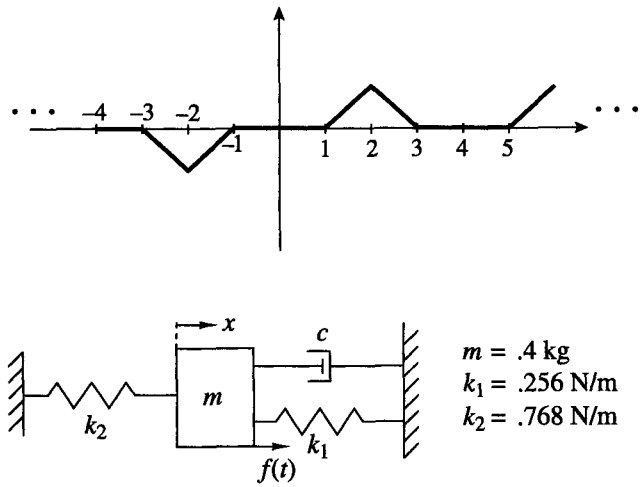


Figure P3.9

3.10. For Figure P3.10, determine whether the coefficients b_m and a_m are zero or nonzero if the function is approximated by

$$f(t) = \sum_{m=0}^{n} a_m \cos(m\omega_0 t) + \sum_{m=1}^{n} b_m \sin(m\omega_0 t)$$

What is the actual value of ω_0 and a_0?

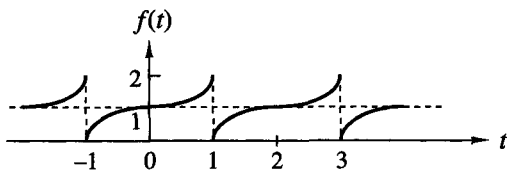


Figure P3.10

3.11. Why don't we retain an infinite number of terms in our Fourier series analyses? Give two reasons. What is a reasonable criterion for deciding how many terms to include our series?

3.12. Use a Fourier series analysis to plot out approximations to the function shown in Figure P3.12. Show how the approximation changes as you go from one included harmonic to several.

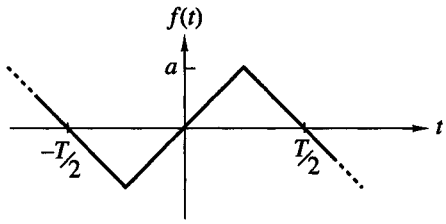


Figure P3.12

3.13. Determine the response of a direct force excited, spring-mass system to the given input. Two cases need to be examined: $T = 3$ seconds and $T = 9.24$ seconds. Look at the harmonic content of the input and the harmonic content of the output and use this information to explain why you obtained qualitatively different responses for the two cases. $m = 5$ kg and $k = 20$ N/m. $f(t)$ given in Figure P3.12.

3.14. Express the following function in terms of a Fourier series representation (illustrated in Figure P3.14). It might model the force input of a pair of woodpeckers located on opposite sides of a tree—sharp, regularly spaced impulses. What difficulties will the Fourier series analysis have with such a function?

$$f(t) = \sum_{n=-\infty}^{\infty} (-1)^{n+1}\delta(t - (2n-1)T)$$

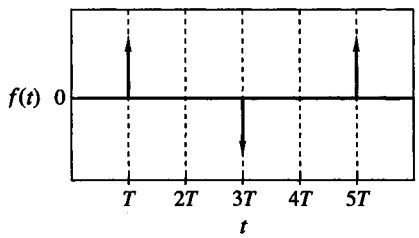


Figure P3.14

Section 3.3

3.15. Determine $x(2.)$ and $\dot{x}(2)$ for a direct force excited, spring-mass system for the input given in Figure P3.15 via a convolution approach. $x(0) = \dot{x}(0) = 0$, $m = 1$ kg, $k = 9$ N/m.

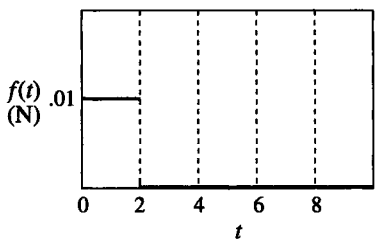


Figure P3.15

3.16. Consider a direct force excited, spring-mass system. Show via a convolution approach that the displacement at $t = .5T$ will be twice the static equilibrium displacement if the force input is constant from $t = 0$ to $t = .5T$.

3.17. Use a convolution approach to determine the response of a seismically excited spring-mass system to a step seismic input.

3.18. Will the force input illustrated in Figure P3.18 lead to a stable oscillation (no growth in amplitude) for a direct force excited, spring-mass system for which $m = 100$ kg and $k = 2500\pi^2$ N/m? Use a convolution approach.

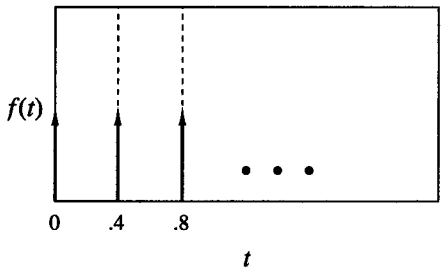


Figure P3.18

Section 3.4

3.19. Determine the maximal acceleration that the mass will experience if dropped onto the spring illustrated in Figure P3.19. $m = 20$ kg, $k = 20,000$ N/m, $h = 1$ m, $g = 9.81$ m/s^2.

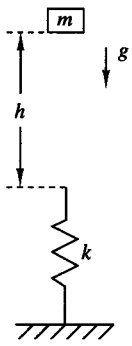


Figure P3.19

3.20. What is the accelerative loading on a mass that's uniformly decelerated from 35 mph to zero in .1 second?

3.21. By adding a crush zone to a car, the time for the occupants' velocity to uniformly decelerate from 60 mph to zero is raised from .07 second to .28 second. By how much does this reduce the deceleration felt by the driver?

3.22. By how much will the maximal acceleration of Problem 3.19 change if the gravitational acceleration is reduced by a factor of 6 (as would happen if one ran the experiment on Earth and then on the moon)?

3.23. In an egg drop contest, the goal is to drop an egg into a container from the greatest height possible without having the egg break. Assume that the egg doesn't break if dropped from a 2 m height in a container that has a stiffness of 400 N/m. The mass of the egg is .05 kg. How must the supporting stiffness be changed to double the height from which the egg can successfully be dropped?

3.24. What is the maximum acceleration felt by a lumped mass that falls onto an arbitrarily stiff surface?

3.25. Will reducing an item's mass decrease the maximal acceleration it will experience if dropped onto an elastic surface?

3.26. A shipping company has bought new trucks for which the height of the cargo bed has been reduced from 1.1 m to .9 m. Unfortunately for the consumer, this company unloads its cargo by dropping it off the cargo bed onto the ground. Model the package as a spring-mass system like the one in Figure P3.26. If $k = 1000$ N/m and $m = 10$ kg, determine how much the maximal accelerative loading will decrease because of the new cargo bed height.

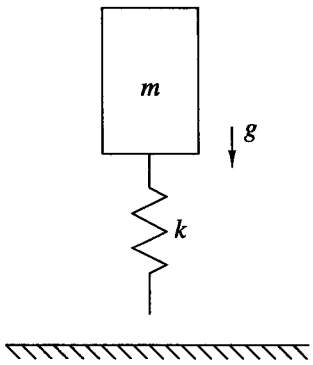


Figure P3.26

3.27. If the drop height of a spring-mass system like the one shown in Figure P3.16 is increased by 20%, by how much must the system's natural frequency be changed for there to be no overall change in the maximum accelerative loading?